

Acne

Symptoms Acne typically appears as a mixture of comedones (blackheads and whiteheads), inflamed papules and pustules, and in more severe cases nodules or cysts. Lesions most often occur on the face, chest, and back where sebaceous glands are abundant. Affected skin may be oily and tender; severe inflammatory acne can be painful and may lead to permanent scarring and post-inflammatory hyperpigmentation. Psychological effects are common and include embarrassment, social withdrawal, anxiety, and lowered self-esteem, particularly in adolescents and young adults.

Treatment Treatment is staged according to severity. For mild acne, first-line therapy is topical agents: benzoyl peroxide to reduce bacteria and comedone formation; topical retinoids (tretinoin, adapalene) to normalize follicular epithelial desquamation; and topical antibiotics combined with benzoyl peroxide to reduce resistance. Moderate inflammatory acne often requires oral antibiotics (doxycycline, minocycline) for their anti-inflammatory and antibacterial effects, usually combined with topical retinoids. Hormonal therapy (combined oral contraceptives, spironolactone) is effective for women with signs of androgen excess or cyclical flares. Severe nodulocystic acne is treated with oral isotretinoin under specialist supervision because of its teratogenicity and systemic side effects; monitoring includes liver function tests and lipid profiles. Adjunctive measures include gentle skin care, avoidance of comedogenic cosmetics, sun protection, and, for scarring, dermatologic procedures such as chemical peels, microneedling, or laser resurfacing. Long-term management focuses on preventing scarring and addressing psychosocial impact.

Addiction (Substance Use Disorders)

Symptoms Addiction is characterized by compulsive use of a substance or engagement in a behavior despite harmful consequences. Core features include intense cravings, impaired control over use, tolerance (needing more to achieve the same effect), and withdrawal symptoms on cessation. Physical signs vary by substance: alcohol misuse may cause tremor, nausea, and autonomic instability; opioid dependence may present with pinpoint pupils, constipation, and sedation; stimulant misuse can cause weight loss, insomnia, and cardiovascular symptoms. Behavioral and social consequences include neglect of responsibilities, relationship breakdown, legal or financial problems, and continued use despite awareness of harm. Co-occurring psychiatric disorders (depression, anxiety, PTSD) are common and complicate treatment.

Treatment Effective treatment is multimodal and individualized. Initial medical stabilization (detoxification) manages acute withdrawal under supervision; benzodiazepines are used for alcohol withdrawal, and opioid withdrawal may be managed with buprenorphine or methadone. Pharmacotherapies reduce cravings and relapse risk: naltrexone and acamprosate for alcohol use disorder; buprenorphine,

methadone, and extended-release naltrexone for opioid use disorder. Psychosocial interventions are central: cognitive behavioral therapy (CBT) addresses maladaptive thoughts and coping skills; motivational interviewing enhances readiness to change; contingency management uses incentives to reinforce abstinence. Residential rehabilitation programs provide structured environments for those with severe dependence. Long-term recovery often requires ongoing counseling, peer support groups (e.g., Alcoholics Anonymous), relapse prevention planning, and treatment of co-occurring mental health conditions. Family involvement and social support improve outcomes.

Attention-Deficit/Hyperactivity Disorder (ADHD)

Symptoms ADHD presents with a persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with functioning. Inattention symptoms include difficulty sustaining attention, careless mistakes, poor organization, forgetfulness, and avoidance of tasks requiring sustained mental effort. Hyperactivity-impulsivity includes fidgeting, inability to remain seated, excessive talking, interrupting others, and acting without considering consequences. Symptoms begin in childhood but may persist into adulthood, where inattentive symptoms and executive dysfunction (time management, planning) often predominate. ADHD frequently coexists with learning disorders, anxiety, and mood disorders.

Treatment Treatment combines pharmacologic and nonpharmacologic strategies. First-line medications are stimulant agents (methylphenidate, amphetamine salts) that increase catecholamine activity and improve attention and impulse control; they are effective in the majority of patients but require monitoring for appetite suppression, sleep disturbance, and cardiovascular effects. Nonstimulant options (atomoxetine, guanfacine, clonidine) are alternatives when stimulants are ineffective or contraindicated. Behavioral therapies include parent training, classroom interventions, organizational skills training, and CBT for adolescents and adults. Educational accommodations (IEP/504 plans) support academic performance. A comprehensive approach addresses comorbid conditions, sleep hygiene, and lifestyle factors such as regular exercise and structured routines.

AIDS (Acquired Immunodeficiency Syndrome)

Symptoms AIDS is the advanced stage of infection with human immunodeficiency virus (HIV), marked by severe immunosuppression and susceptibility to opportunistic infections and certain malignancies. Early HIV infection may be asymptomatic or present with a mononucleosis-like illness. As immune function declines, patients develop recurrent infections (pneumocystis pneumonia, candidiasis), chronic diarrhea, weight loss, persistent fevers, night sweats, generalized lymphadenopathy, and neurologic complications. AIDS-defining illnesses include certain opportunistic

infections and cancers such as Kaposi sarcoma and non-Hodgkin lymphoma. Laboratory markers include progressive CD4+ T-cell decline and high viral load.

Treatment Antiretroviral therapy (ART) is the cornerstone of HIV care and transforms HIV into a manageable chronic disease when started early and taken consistently. Modern ART regimens combine drugs from different classes (e.g., integrase inhibitors, nucleoside reverse transcriptase inhibitors) to suppress viral replication, restore immune function, and reduce transmission risk. Treatment is individualized based on resistance testing, comorbidities, and drug interactions. Management also includes prophylaxis and treatment of opportunistic infections, vaccination where appropriate, screening for and treating co-infections (hepatitis B/C, tuberculosis), and addressing psychosocial needs. Regular monitoring of CD4 counts and viral load guides therapy. Prevention strategies include pre-exposure prophylaxis (PrEP) for high-risk individuals and post-exposure prophylaxis (PEP) after potential exposure.

Alcoholism (Alcohol Use Disorder)

Symptoms Alcohol use disorder ranges from hazardous drinking to severe dependence. Symptoms include inability to limit drinking, craving, continued use despite harm, tolerance, and withdrawal symptoms (tremor, anxiety, nausea, seizures in severe cases). Chronic heavy drinking leads to medical complications: liver disease (fatty liver, hepatitis, cirrhosis), cardiomyopathy, neuropathy, pancreatitis, and increased cancer risk. Cognitive impairment and mood disorders are common. Social and occupational dysfunction, legal problems, and family disruption frequently accompany the disorder.

Treatment Treatment begins with assessment of severity and medical stabilization for withdrawal; benzodiazepines are commonly used to prevent severe withdrawal complications. Pharmacologic options to support abstinence include naltrexone (reduces craving and reward), acamprosate (supports abstinence), and disulfiram (produces aversive reaction to alcohol). Psychosocial interventions—CBT, motivational enhancement therapy, and mutual-help groups—are effective components of care. Integrated treatment for co-occurring psychiatric disorders improves outcomes. For advanced liver disease, referral for transplant evaluation may be necessary after sustained abstinence. Long-term follow-up, relapse prevention planning, and social support are essential.

Allergies (Allergic Rhinitis, Food and Drug Allergies, Atopy)

Symptoms Allergic conditions manifest when the immune system reacts to otherwise harmless substances (allergens). Common symptoms include sneezing, nasal congestion, rhinorrhea, itchy and watery eyes, skin rashes (urticaria, eczema), and in severe cases angioedema or anaphylaxis. Food allergies can cause gastrointestinal symptoms, hives, and anaphylaxis. Drug allergies may present with rashes, fever, or systemic reactions. Allergic asthma presents with wheeze, cough, and breathlessness.

triggered by allergens. Symptoms may be seasonal or perennial depending on the allergen.

Treatment Management includes allergen avoidance when feasible, pharmacotherapy, and immunotherapy. For allergic rhinitis, intranasal corticosteroids are the most effective controller therapy; oral or intranasal antihistamines relieve itching and sneezing. Leukotriene receptor antagonists can help in allergic asthma and rhinitis. For acute urticaria or mild allergic reactions, oral antihistamines and topical corticosteroids are used; anaphylaxis requires immediate intramuscular epinephrine and emergency care. Allergen immunotherapy (subcutaneous or sublingual) is effective for long-term reduction of sensitivity to inhalant allergens and can reduce medication needs. For food allergies, strict avoidance and education on emergency epinephrine use are essential. Identification of triggers through history and testing (skin prick, specific IgE) guides management.

Alzheimer's Disease

Symptoms Alzheimer's disease is a progressive neurodegenerative disorder characterized by gradual decline in memory, executive function, language, and visuospatial skills. Early signs include short-term memory loss and difficulty learning new information; as disease advances, patients develop disorientation, impaired judgment, behavioral changes (apathy, agitation, wandering), and loss of ability to perform activities of daily living. Neuropsychiatric symptoms such as depression, psychosis, and sleep disturbances are common. The course is progressive over years, ultimately leading to severe cognitive impairment and dependence.

Treatment There is no cure, but treatments can modestly slow symptom progression and improve quality of life. Cholinesterase inhibitors (donepezil, rivastigmine, galantamine) are used in mild to moderate disease to enhance cholinergic neurotransmission and may improve cognition and function. Memantine, an NMDA receptor antagonist, is used in moderate to severe stages to reduce excitotoxicity. Nonpharmacologic interventions—cognitive stimulation, structured routines, environmental modifications, and caregiver education—are critical. Management of behavioral symptoms may include behavioral strategies first; antipsychotics are reserved for severe agitation or psychosis due to risks. Addressing comorbid medical conditions, ensuring safety, and planning for long-term care needs are essential components of care.

Anemia

Symptoms Anemia is a reduction in hemoglobin concentration or red blood cell mass, producing symptoms related to decreased oxygen delivery: fatigue, weakness, pallor, shortness of breath on exertion, palpitations, and dizziness. Severe or rapidly developing anemia can cause chest pain, syncope, or heart failure. Specific types of

anemia may have additional signs: iron-deficiency anemia can cause pica and brittle nails; hemolytic anemias may present with jaundice and dark urine; megaloblastic anemia (B12/folate deficiency) can cause neurologic symptoms such as paresthesias and gait disturbance.

Treatment Treatment targets the underlying cause. Iron-deficiency anemia is treated with oral iron supplementation and investigation for sources of blood loss (gastrointestinal bleeding, heavy menstruation); intravenous iron is used when oral therapy is not tolerated or absorption is impaired. Vitamin B12 deficiency requires parenteral or high-dose oral B12 replacement; folate deficiency is treated with folic acid. Hemolytic anemias may require immunosuppression for autoimmune causes, transfusions for severe anemia, or disease-specific therapies (e.g., splenectomy in select hereditary hemolytic disorders). Chronic disease-related anemia management focuses on treating the underlying illness and may include erythropoiesis-stimulating agents in selected patients with renal disease or cancer. Monitoring response with hemoglobin and reticulocyte counts guides therapy.

Anorexia Nervosa

Symptoms Anorexia nervosa is an eating disorder marked by self-imposed starvation, intense fear of weight gain, and distorted body image. Patients severely restrict caloric intake, may engage in excessive exercise, purging, or laxative misuse, and often have a significantly low body weight relative to age and sex. Medical complications are widespread: bradycardia, hypotension, electrolyte disturbances, amenorrhea, osteoporosis, gastrointestinal dysmotility, and increased mortality risk from cardiac arrhythmia or suicide. Psychological features include perfectionism, denial of illness, and comorbid depression or anxiety.

Treatment Treatment is multidisciplinary and urgent when medical instability is present. Initial goals are medical stabilization and nutritional rehabilitation, often requiring inpatient care for severe malnutrition or electrolyte abnormalities. Refeeding must be carefully managed to avoid refeeding syndrome. Psychotherapeutic approaches include family-based therapy (especially for adolescents), cognitive behavioral therapy, and supportive psychotherapy to address distorted beliefs about weight and food. Pharmacotherapy has limited benefit for core anorexia symptoms but may treat comorbid depression or anxiety. Long-term follow-up addresses bone health, reproductive issues, and relapse prevention. Collaboration among physicians, psychiatrists, dietitians, and family members is essential.

Anxiety Disorders

Symptoms Anxiety disorders encompass generalized anxiety disorder (GAD), panic disorder, social anxiety disorder, specific phobias, and others. Core symptoms include excessive worry, restlessness, muscle tension, sleep disturbance, irritability, and

difficulty concentrating. Panic disorder features recurrent unexpected panic attacks—sudden surges of intense fear with palpitations, sweating, trembling, shortness of breath, and fear of losing control. Social anxiety causes marked fear of social situations and avoidance. Anxiety often coexists with depression and substance use disorders and can cause significant functional impairment.

Treatment Evidence-based treatment combines psychotherapy and pharmacotherapy. Cognitive behavioral therapy (CBT) is first-line for many anxiety disorders and teaches cognitive restructuring, exposure techniques, and coping skills. Pharmacologic options include selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) as first-line medications; benzodiazepines provide rapid symptom relief but are used cautiously due to dependence risk. For panic disorder, CBT with interoceptive exposure plus an SSRI is effective. Social anxiety benefits from CBT with exposure and sometimes beta-blockers for performance situations. Adjunctive strategies include relaxation training, mindfulness, sleep hygiene, and addressing comorbid conditions. Treatment is individualized and often continued for months to years to prevent relapse.

Back Pain

Symptoms Back pain ranges from a dull, constant ache to a sudden, sharp pain that leaves a person incapacitated. It may be localized to the lower back (lumbar), mid-back (thoracic), or upper back (cervical), and can radiate into the buttocks, legs (sciatica), or arms depending on nerve involvement. Symptoms include muscle stiffness, limited range of motion, difficulty standing or walking, and pain that worsens with certain movements or prolonged sitting. Acute back pain often follows an injury or heavy lifting; chronic back pain persists for months and may be associated with degenerative changes, spinal stenosis, or disc disease. Red flags such as fever, unexplained weight loss, progressive neurological deficits (numbness, weakness, bowel/bladder dysfunction) require urgent evaluation.

Treatment Initial management emphasizes conservative care: relative rest for a short period, heat or cold packs, and gradual return to activity. Analgesics such as acetaminophen or nonsteroidal anti-inflammatory drugs (NSAIDs) relieve pain and inflammation; short courses of muscle relaxants may help with severe spasm. Physical therapy is central to recovery, focusing on core strengthening, flexibility, posture correction, and ergonomic education to prevent recurrence. For radicular pain due to nerve root compression, short courses of oral corticosteroids or epidural steroid injections may reduce inflammation and pain. Chronic or refractory cases may benefit from multidisciplinary pain management including cognitive behavioral therapy, graded exercise programs, and interventional procedures (nerve blocks, radiofrequency ablation). Surgical options (discectomy, laminectomy, spinal fusion) are reserved for structural causes with persistent neurological deficits or intractable pain despite

conservative therapy. Prevention includes weight management, regular exercise, proper lifting techniques, and workplace ergonomics.

Bipolar Disorder

Symptoms Bipolar disorder is characterized by mood episodes that alternate between mania or hypomania and depression. Manic episodes involve abnormally elevated or irritable mood, increased energy, decreased need for sleep, grandiosity, pressured speech, racing thoughts, impulsive or risky behaviors, and impaired judgment; severe mania can include psychosis. Hypomania is a milder form that does not cause marked impairment. Depressive episodes feature persistent low mood, loss of interest, fatigue, changes in appetite or sleep, feelings of worthlessness, and suicidal ideation. The pattern and frequency of episodes vary: some patients have long periods of euthymia, while others experience rapid cycling. Bipolar disorder often begins in late adolescence or early adulthood and can significantly impair social and occupational functioning.

Treatment Treatment aims to stabilize mood, prevent relapse, and manage acute episodes. Mood stabilizers such as lithium remain a cornerstone for long-term maintenance and reduce suicide risk; therapeutic monitoring of blood levels and renal/thyroid function is required. Anticonvulsants (valproate, carbamazepine, lamotrigine) are used as mood stabilizers, particularly for rapid cycling or mixed features. Atypical antipsychotics (quetiapine, olanzapine, risperidone) are effective for acute mania and as maintenance therapy. For acute mania, hospitalization may be necessary for safety and stabilization. Depressive episodes are treated cautiously; some antidepressants can trigger mania and are usually combined with a mood stabilizer. Psychotherapy (psychoeducation, family therapy, CBT) supports medication adherence, early recognition of relapse signs, and coping strategies. Lifestyle measures—regular sleep, routine, stress management, and substance avoidance—are important. Long-term follow-up and coordination with mental health services are essential.

Bladder Infections (Cystitis / Urinary Tract Infection)

Symptoms Bladder infections typically present with urinary urgency, frequency, dysuria (painful or burning urination), suprapubic discomfort, and sometimes hematuria (blood in urine). In uncomplicated lower urinary tract infections, systemic symptoms are usually absent; fever and flank pain suggest upper tract involvement (pyelonephritis). Elderly patients may present atypically with confusion, falls, or generalized weakness. Recurrent infections are common in some populations and may indicate anatomic abnormalities, incomplete bladder emptying, or other risk factors such as catheter use or diabetes.

Treatment Uncomplicated cystitis is commonly treated with short courses of oral antibiotics guided by local resistance patterns (e.g., nitrofurantoin,

trimethoprim-sulfamethoxazole, fosfomycin). Symptomatic relief with analgesics (phenazopyridine for short term) and increased fluid intake is helpful. For recurrent UTIs, evaluation for predisposing factors is recommended; prophylactic strategies include behavioral measures (urinating after intercourse, hydration), topical vaginal estrogen for postmenopausal women, or low-dose prophylactic antibiotics in selected cases. Complicated UTIs, pyelonephritis, or infections in pregnant women require broader antibiotic coverage and sometimes hospitalization for intravenous therapy. Catheter-associated infections require catheter management and targeted antibiotics. Culture and sensitivity testing guide therapy in recurrent or complicated cases.

Blood Pressure (Hypertension)

Symptoms Hypertension is often asymptomatic and is frequently detected during routine screening. When symptoms occur, they are nonspecific: headaches (especially occipital), dizziness, visual disturbances, or palpitations. Severe or malignant hypertension can present with chest pain, shortness of breath, neurological deficits, or signs of end-organ damage (retinopathy, renal impairment). Long-standing uncontrolled hypertension increases the risk of heart disease, stroke, kidney failure, and peripheral vascular disease.

Treatment Management combines lifestyle modification and pharmacotherapy. Lifestyle measures include dietary sodium reduction, the DASH diet (rich in fruits, vegetables, low-fat dairy), weight loss, regular aerobic exercise, moderation of alcohol intake, and smoking cessation. First-line antihypertensive drug classes include thiazide diuretics, ACE inhibitors, angiotensin receptor blockers (ARBs), calcium channel blockers, and beta-blockers (in selected patients). Choice of agent depends on comorbidities (diabetes, chronic kidney disease, heart failure), age, and race. Treatment goals are individualized but generally aim for sustained blood pressure control to reduce cardiovascular risk. Regular monitoring, adherence support, and management of secondary causes (renal artery stenosis, endocrine disorders) are part of comprehensive care. In hypertensive emergencies with acute end-organ damage, immediate hospitalization and controlled blood pressure reduction with intravenous agents are required.

Bone Fractures

Symptoms A fracture typically presents with acute localized pain, swelling, deformity, bruising, and inability to bear weight or use the affected limb. Crepitus or abnormal mobility may be present. Open fractures involve skin disruption and carry infection risk. Certain fractures (hip, vertebral) may present with more subtle symptoms in elderly patients, such as inability to walk or back pain with height loss. Neurovascular compromise (numbness, absent pulses) indicates urgent evaluation.

Treatment Initial management follows trauma principles: immobilization, pain control, and assessment for neurovascular injury. Closed fractures are often managed with reduction (realignment) and immobilization using casts or splints. Some fractures require surgical fixation with plates, screws, intramedullary nails, or external fixation to restore alignment and allow early mobilization. Vertebral compression fractures may be treated conservatively with analgesia and bracing; vertebroplasty or kyphoplasty are options in select cases. Open fractures require urgent irrigation, debridement, and antibiotics to prevent infection. Rehabilitation with physiotherapy is essential to restore function and strength. Prevention strategies include fall risk assessment, bone health optimization (calcium, vitamin D, bisphosphonates for osteoporosis), and safety measures.

Breast Cancer

Symptoms Breast cancer commonly presents as a new, painless lump in the breast or axilla, but symptoms vary: skin changes (dimpling, peau d'orange), nipple retraction, nipple discharge (especially bloody), or localized pain. Advanced disease may cause systemic symptoms such as weight loss, bone pain, or respiratory symptoms from metastases. Screening mammography detects many asymptomatic cancers at an earlier stage. Risk factors include age, family history, genetic mutations (BRCA1/2), hormonal factors, and certain lifestyle elements.

Treatment Treatment is multimodal and tailored to tumor stage, receptor status (ER/PR/HER2), and patient factors. Local therapy includes breast-conserving surgery (lumpectomy) with sentinel lymph node biopsy or mastectomy, often followed by radiation therapy for local control. Systemic therapy includes endocrine therapy for hormone receptor-positive tumors (tamoxifen, aromatase inhibitors), chemotherapy for high-risk or triple-negative cancers, and targeted agents for HER2-positive disease (trastuzumab, pertuzumab). Neoadjuvant therapy may shrink tumors preoperatively. Supportive care addresses lymphedema prevention, psychosocial needs, and survivorship issues. Multidisciplinary care involving surgical, medical, and radiation oncology optimizes outcomes. Genetic counseling is recommended for patients with strong family histories or early-onset disease.

Bronchitis (Acute and Chronic)

Symptoms Acute bronchitis typically follows an upper respiratory infection and presents with cough (initially dry, later productive), chest discomfort, wheeze, and sometimes low-grade fever. Symptoms usually last several weeks. Chronic bronchitis, a form of chronic obstructive pulmonary disease (COPD), is defined by productive cough for at least three months in two consecutive years and is associated with smoking. Chronic bronchitis leads to persistent cough, sputum production, recurrent infections, and progressive airflow limitation.

Treatment Acute bronchitis is usually viral and managed with symptomatic care: hydration, rest, antitussives for severe cough, and bronchodilators for wheeze. Antibiotics are not routinely indicated unless bacterial infection is suspected or in patients with chronic lung disease. For chronic bronchitis, smoking cessation is the most important intervention to slow disease progression. Bronchodilators (short- and long-acting beta-agonists, anticholinergics) and inhaled corticosteroids for selected patients improve symptoms and reduce exacerbations. Pulmonary rehabilitation, vaccination (influenza, pneumococcal), and oxygen therapy for advanced disease are part of comprehensive care. Management of acute exacerbations may require antibiotics, systemic corticosteroids, and hospitalization if severe.

Burns

Symptoms Burn injuries vary by depth and extent. Superficial (first-degree) burns cause erythema and pain without blistering. Partial-thickness (second-degree) burns produce blisters, intense pain, and moist appearance. Full-thickness (third-degree) burns destroy skin layers, may appear white or charred, and can be painless due to nerve destruction. Large or deep burns risk fluid loss, hypothermia, infection, and systemic inflammatory response. Inhalation injury from smoke can cause airway compromise and respiratory failure. Burn severity is assessed by depth and percentage of total body surface area (TBSA) involved.

Treatment Initial care follows ABCs (airway, breathing, circulation). For significant burns, fluid resuscitation using formulas (Parkland) prevents hypovolemia. Pain control, wound care (cleansing, debridement), and topical antimicrobials reduce infection risk. Superficial burns heal with conservative care; deeper burns may require surgical excision and skin grafting. Tetanus prophylaxis and early nutritional support are important. Severe burns are best managed in specialized burn centers with multidisciplinary teams addressing wound care, infection control, rehabilitation, and psychological support. Prevention of contractures and long-term scar management (pressure garments, laser therapy) are part of recovery.

Bulimia Nervosa

Symptoms Bulimia nervosa is an eating disorder characterized by recurrent episodes of binge eating followed by compensatory behaviors to prevent weight gain, such as self-induced vomiting, laxative or diuretic misuse, fasting, or excessive exercise. Patients often maintain a normal weight, making the disorder less obvious. Medical complications include electrolyte disturbances (hypokalemia, hyponatremia), dehydration, dental erosion, esophagitis, gastrointestinal problems, and cardiac arrhythmias. Psychological features include shame, secrecy, and comorbid mood or anxiety disorders.

Treatment Treatment combines psychotherapy, nutritional rehabilitation, and medical monitoring. Cognitive behavioral therapy (CBT) is the first-line psychotherapeutic approach and focuses on interrupting the binge-purge cycle and addressing distorted body image. Interpersonal therapy and family-based approaches may also help. Pharmacotherapy with SSRIs (fluoxetine) can reduce binge frequency and treat comorbid depression or anxiety. Medical management addresses electrolyte imbalances and complications; hospitalization may be necessary for severe electrolyte disturbances or cardiac risk. Long-term follow-up and relapse prevention strategies are essential.

Brain Tumors

Symptoms Symptoms of brain tumors depend on tumor location, size, and growth rate. Common presentations include persistent headaches (often worse in the morning), seizures, focal neurological deficits (weakness, sensory loss, speech or vision changes), cognitive or personality changes, nausea, and vomiting. Increased intracranial pressure can cause progressive headache, papilledema, and altered consciousness. Some tumors produce endocrine disturbances if they involve the pituitary region. Symptoms may be subtle and progress gradually, leading to delayed diagnosis.

Treatment Management is multidisciplinary and tailored to tumor type (primary vs metastatic), histology, location, and patient factors. Surgical resection is often the first step to obtain tissue diagnosis and reduce mass effect; the extent of resection balances tumor removal with preservation of neurological function. Radiation therapy (conventional or stereotactic radiosurgery) treats residual disease or inoperable tumors. Chemotherapy is used for certain tumor types (glioblastoma, medulloblastoma) and targeted therapies are emerging for tumors with specific molecular markers. Supportive care includes anticonvulsants for seizure control, corticosteroids to reduce edema, and rehabilitation for functional deficits. For metastatic brain disease, systemic cancer control and local therapies (surgery, radiosurgery) are coordinated with oncology. Prognosis varies widely by tumor type and stage.

Cancer

Symptoms Cancer is not a single disease but a group of disorders characterized by uncontrolled cell growth; symptoms vary widely by cancer type and stage. Common early signs can be subtle and nonspecific: unexplained weight loss, persistent fatigue, night sweats, and low-grade fevers. Local symptoms depend on the organ involved — for example, a lung tumor may cause a persistent cough, hemoptysis (coughing blood), or breathlessness; a colorectal tumor may cause changes in bowel habits, rectal bleeding, or abdominal pain; a breast tumor may present as a new lump, nipple

discharge, or skin changes. Advanced disease often produces systemic symptoms such as cachexia (wasting), bone pain from metastases, neurological deficits from brain involvement, or jaundice from liver metastases. Because early symptoms can mimic benign conditions, persistent or progressive signs warrant medical evaluation.

Treatment Cancer treatment is individualized and often multimodal, combining local and systemic therapies. Local control is achieved with surgery to remove tumors when feasible and radiation therapy to eradicate residual disease or palliate symptoms. Systemic therapies include chemotherapy, which uses cytotoxic agents to target rapidly dividing cells; targeted therapies that inhibit specific molecular pathways driving tumor growth (e.g., tyrosine kinase inhibitors); hormonal therapies for hormone-sensitive cancers (breast, prostate); and immunotherapies that harness the immune system to attack cancer cells (checkpoint inhibitors, CAR-T in select settings). Treatment selection depends on tumor histology, stage, molecular markers, patient performance status, and goals of care. Supportive care addresses pain control, management of treatment side effects (nausea, neutropenia, mucositis), nutritional support, and psychosocial needs. Palliative care is integrated early for symptom relief and quality-of-life optimization. Long-term follow-up includes surveillance for recurrence and management of late effects of therapy.

Cataracts

Symptoms Cataracts are an opacity of the eye's crystalline lens that develops gradually, most commonly with aging. Early symptoms include blurred or cloudy vision, increased glare or halos around lights (especially at night), difficulty reading small print, faded or yellowed colors, and the need for more light to see clearly. Patients may notice frequent changes in eyeglass prescription or improved near vision (a temporary "second sight" effect) as the lens changes. When cataracts progress, they significantly impair daily activities such as driving and reading and increase fall risk in older adults.

Treatment The only definitive treatment for visually significant cataracts is surgical removal of the cloudy lens and implantation of an artificial intraocular lens (IOL). Cataract surgery is one of the most common and successful procedures worldwide and is typically performed on an outpatient basis under local anesthesia. Preoperative assessment includes visual acuity testing, slit-lamp examination, and biometry to select the appropriate IOL power. For patients with mild cataracts, initial management focuses on optimizing lighting, using magnification aids, and updating eyeglass prescriptions. Surgery is recommended when visual impairment affects quality of life or safety. Postoperative care includes topical antibiotics and anti-inflammatory drops, activity modifications for a short period, and follow-up to monitor healing and visual outcomes.

Celiac Disease

Symptoms Celiac disease is an immune-mediated enteropathy triggered by gluten ingestion in genetically susceptible individuals. Classic symptoms include chronic diarrhea, steatorrhea (fatty stools), abdominal pain, bloating, and weight loss due to malabsorption. Nonclassic or extraintestinal presentations are common and include iron-deficiency anemia, fatigue, osteoporosis from calcium and vitamin D malabsorption, dermatitis herpetiformis (itchy blistering skin rash), neurologic symptoms (peripheral neuropathy, ataxia), and reproductive issues (infertility, recurrent miscarriage). Some patients have minimal gastrointestinal complaints but present with subtle signs such as short stature in children or unexplained elevated liver enzymes.

Treatment The cornerstone of treatment is a strict, lifelong gluten-free diet, eliminating wheat, barley, and rye to allow intestinal healing and prevent complications. Dietary adherence typically leads to symptom resolution and mucosal recovery over months to years; nutritional deficiencies (iron, folate, B12, vitamin D) should be identified and corrected. Patients benefit from dietitian support for label reading, cross-contamination avoidance, and balanced nutrition. In refractory cases or when symptoms persist despite a gluten-free diet, evaluation for inadvertent gluten exposure, alternative diagnoses, or refractory celiac disease is necessary; immunosuppressive therapies (e.g., corticosteroids) may be used in severe refractory disease. Regular follow-up includes symptom assessment, serologic testing for antibodies, and monitoring for complications such as osteoporosis and associated autoimmune conditions.

Cerebral Palsy

Symptoms Cerebral palsy (CP) is a group of permanent movement and posture disorders caused by nonprogressive disturbances in the developing fetal or infant brain. Symptoms vary widely in type and severity: spasticity (stiff, tight muscles), dyskinesia (involuntary movements), ataxia (poor coordination), or mixed patterns. Motor impairments lead to delayed milestones, abnormal gait, difficulty with fine motor tasks, and impaired balance. Associated problems commonly include intellectual disability, speech and language difficulties, visual or hearing impairments, epilepsy, and feeding/swallowing problems. The clinical picture is established in early childhood but the functional impact persists lifelong.

Treatment There is no cure for CP, but early, multidisciplinary interventions maximize function and quality of life. Physical therapy focuses on stretching, strengthening, gait training, and use of orthoses to improve mobility and prevent contractures. Occupational therapy enhances fine motor skills and adaptive strategies for daily living. Speech and language therapy addresses communication and swallowing. Pharmacologic treatments reduce spasticity and pain: oral agents (baclofen, tizanidine), botulinum toxin injections for focal spasticity, and intrathecal baclofen pumps for severe generalized spasticity. Orthopedic surgeries (tendon lengthening, osteotomies) correct deformities and improve function in selected patients.

Management of seizures, nutritional support, assistive devices (wheelchairs, communication aids), and educational and social support are integral. Lifelong care planning and transition to adult services are important for long-term outcomes.

Chickenpox (Varicella)

Symptoms Chickenpox, caused by varicella-zoster virus, typically presents with a prodrome of fever, malaise, and headache followed by a characteristic pruritic vesicular rash. Lesions appear in successive crops and are found on the trunk, face, and scalp, often spreading to the extremities; lesions progress from macules to papules to vesicles and then crust over. The rash is intensely itchy and may be more severe in adolescents and adults than in children. Complications include bacterial superinfection of lesions, pneumonia (particularly in adults and immunocompromised patients), encephalitis, and, rarely, hemorrhagic varicella.

Treatment Management is largely supportive for healthy children: antipyretics (acetaminophen), skin care to reduce scratching (cool baths, calamine lotion), and antihistamines for pruritus. Aspirin should be avoided in children due to the risk of Reye syndrome. Antiviral therapy with acyclovir is recommended for adolescents, adults, pregnant women, and immunocompromised patients to reduce severity and complications; it is most effective when started within 24–48 hours of rash onset. Varicella vaccination is the primary preventive measure and has dramatically reduced disease incidence and complications; post-exposure prophylaxis with varicella vaccine or varicella-zoster immune globulin is considered for high-risk contacts. Hospitalization is required for severe complications such as varicella pneumonia or encephalitis.

Cholesterol (Hyperlipidemia)

Symptoms High cholesterol itself is usually asymptomatic and is typically detected through routine blood tests measuring lipid profiles (total cholesterol, LDL, HDL, triglycerides). Symptoms arise from complications of long-standing hyperlipidemia: atherosclerotic cardiovascular disease manifests as chest pain (angina), myocardial infarction, transient ischemic attacks or stroke, and peripheral arterial disease causing claudication. Physical signs of severe lipid disorders may include xanthomas (cholesterol deposits in skin or tendons) and corneal arcus in younger patients.

Treatment Management aims to reduce cardiovascular risk through lifestyle modification and pharmacotherapy when indicated. Lifestyle measures include a heart-healthy diet (reduced saturated and trans fats, increased fiber), regular aerobic exercise, weight loss, smoking cessation, and moderation of alcohol. First-line pharmacologic therapy for elevated LDL cholesterol is statins, which reduce cholesterol synthesis and lower cardiovascular events; dosing is based on baseline risk and treatment goals. For patients intolerant to statins or requiring additional LDL lowering, ezetimibe, PCSK9 inhibitors, or bile acid sequestrants may be used. Management of

hypertriglyceridemia includes addressing secondary causes (alcohol, uncontrolled diabetes), fibrates, omega-3 fatty acids, and lifestyle changes. Regular monitoring of lipid levels, liver function tests when on therapy, and individualized risk assessment guide treatment intensity.

Chronic Obstructive Pulmonary Disease (COPD)

Symptoms COPD is a progressive respiratory disease characterized by persistent airflow limitation, most commonly caused by long-term exposure to noxious particles such as cigarette smoke. Symptoms include chronic productive cough, dyspnea on exertion that progressively worsens, wheeze, and frequent respiratory infections. Exacerbations—acute worsening of respiratory symptoms—lead to increased morbidity and may require hospitalization. Advanced disease can cause hypoxemia, hypercapnia, pulmonary hypertension, and right heart failure (cor pulmonale).

Treatment Management includes smoking cessation as the single most effective intervention to slow disease progression. Pharmacologic therapy uses bronchodilators (short- and long-acting beta-agonists and anticholinergics) to relieve symptoms and reduce exacerbations; inhaled corticosteroids are added for patients with frequent exacerbations and significant eosinophilic inflammation. Pulmonary rehabilitation improves exercise tolerance and quality of life. Long-term oxygen therapy is indicated for patients with chronic hypoxemia to improve survival. Vaccinations (influenza, pneumococcal) reduce infection risk. Acute exacerbations are treated with short courses of systemic corticosteroids, antibiotics when bacterial infection is suspected, and increased bronchodilator therapy; severe exacerbations may require noninvasive ventilation or ICU care. End-of-life planning and palliative care are important for advanced COPD.

Cirrhosis

Symptoms Cirrhosis is the end stage of chronic liver disease characterized by fibrosis and regenerative nodules that impair hepatic function and portal blood flow. Early symptoms may be nonspecific—fatigue, anorexia, weight loss—but progressive disease leads to jaundice, pruritus, easy bruising, spider angiomas, palmar erythema, and muscle wasting. Portal hypertension causes splenomegaly, ascites (abdominal fluid accumulation), and variceal bleeding. Hepatic encephalopathy presents with confusion, asterix, and altered consciousness. Complications include spontaneous bacterial peritonitis, hepatorenal syndrome, and hepatocellular carcinoma.

Treatment Treatment focuses on managing the underlying cause, preventing and treating complications, and assessing for liver transplantation in eligible patients. Etiology-specific measures include antiviral therapy for viral hepatitis, abstinence and treatment for alcohol-related liver disease, and metabolic control for nonalcoholic fatty liver disease. Management of portal hypertension and ascites includes sodium

restriction, diuretics (spironolactone, furosemide), large-volume paracentesis for tense ascites, and nonselective beta-blockers or endoscopic variceal ligation to prevent variceal bleeding. Hepatic encephalopathy is treated with lactulose and rifaximin to reduce ammonia levels. Surveillance for hepatocellular carcinoma with ultrasound and alpha-fetoprotein is recommended in cirrhotic patients. Advanced liver failure may require referral for transplantation; supportive care and management of complications are essential while awaiting definitive therapy.

Common Cold (Viral Upper Respiratory Infection)

Symptoms The common cold is an acute, self-limited viral infection of the upper respiratory tract, most often caused by rhinoviruses, coronaviruses, and other respiratory viruses. Symptoms typically begin with nasal congestion, rhinorrhea (runny nose), sneezing, sore throat, and mild cough. Fever is uncommon in adults but may occur in children. Symptoms usually peak within 2–3 days and resolve within 7–10 days, though cough can persist longer. Complications include secondary bacterial sinusitis, otitis media (especially in children), and exacerbation of asthma or COPD.

Treatment Treatment is symptomatic and supportive. Rest, adequate hydration, saline nasal irrigation, and humidification relieve nasal symptoms. Analgesics and antipyretics (acetaminophen, NSAIDs) manage fever and myalgias. Decongestants and intranasal corticosteroids may provide short-term relief for nasal congestion; antihistamines can reduce rhinorrhea in some patients. Antibiotics are not indicated for uncomplicated viral colds and should be reserved for confirmed bacterial complications. Preventive measures include hand hygiene, respiratory etiquette, and avoiding close contact with infected individuals. For high-risk patients, prompt evaluation is warranted if symptoms worsen or systemic signs develop.

COVID-19 (SARS-CoV-2 Infection)

Symptoms COVID-19 presents with a wide spectrum of illness, from asymptomatic infection to severe respiratory failure and multisystem organ dysfunction. Common symptoms include fever, cough, fatigue, myalgias, sore throat, anosmia (loss of smell), ageusia (loss of taste), nasal congestion, and headache. Gastrointestinal symptoms (nausea, vomiting, diarrhea) occur in some patients. Severe disease is characterized by dyspnea, hypoxemia, pneumonia, acute respiratory distress syndrome (ARDS), thromboembolic events, and multi-organ failure. Long COVID refers to persistent symptoms—fatigue, cognitive impairment, dyspnea, and others—that continue for weeks to months after acute infection.

Treatment Management depends on disease severity. Mild cases are managed at home with supportive care: rest, hydration, antipyretics, and monitoring for worsening symptoms. Antiviral therapy (e.g., oral antivirals or remdesivir in hospitalized patients) is indicated for high-risk individuals early in the disease course to reduce progression. For

patients requiring oxygen or hospitalization, corticosteroids (dexamethasone) reduce mortality in those with hypoxemia; anticoagulation is used to prevent thrombotic complications. Monoclonal antibodies and other targeted therapies have been used for early treatment in high-risk patients, though effectiveness depends on circulating variants. Severe cases may require advanced respiratory support (high-flow oxygen, noninvasive ventilation, mechanical ventilation) and critical care management. Vaccination remains the primary prevention strategy, significantly reducing severe disease, hospitalization, and death. Rehabilitation and multidisciplinary care address long COVID symptoms.

Diabetes (Type 1 and Type 2)

Symptoms Diabetes mellitus presents with polyuria (frequent urination), polydipsia (excessive thirst), polyphagia (increased hunger), unexplained weight loss (more common in type 1), fatigue, blurred vision, and slow wound healing. Type 1 diabetes often has abrupt onset in children or young adults and may present with diabetic ketoacidosis (nausea, vomiting, abdominal pain, rapid breathing). Type 2 diabetes develops gradually and is associated with insulin resistance, obesity, and metabolic syndrome; many patients are asymptomatic early and are diagnosed on screening. Chronic hyperglycemia leads to microvascular complications (retinopathy, nephropathy, neuropathy) and macrovascular disease (coronary artery disease, stroke, peripheral arterial disease).

Treatment Management includes lifestyle modification (dietary changes, weight loss, regular physical activity) and pharmacotherapy. Type 1 diabetes requires lifelong insulin therapy delivered by injections or insulin pump, with education on carbohydrate counting, glucose monitoring, and hypoglycemia management. Type 2 diabetes treatment begins with lifestyle measures and metformin; additional agents (SGLT2 inhibitors, GLP-1 receptor agonists, DPP-4 inhibitors, sulfonylureas, insulin) are added based on glycemic control, comorbidities, and cardiovascular/renal benefits. Regular monitoring of HbA1c, self-monitoring of blood glucose, screening for complications, blood pressure and lipid control, and vaccination are essential. Patient education, multidisciplinary care, and individualized targets improve outcomes.

Depression (Major Depressive Disorder)

Symptoms Major depressive disorder is characterized by persistent low mood, loss of interest or pleasure (anhedonia), changes in appetite or weight, sleep disturbances (insomnia or hypersomnia), psychomotor changes, fatigue, feelings of worthlessness or excessive guilt, diminished concentration, and recurrent thoughts of death or suicide. Symptoms must persist for at least two weeks and cause significant impairment. Depression often coexists with anxiety, substance use, and chronic medical conditions, complicating diagnosis and treatment.

Treatment Treatment combines psychotherapy and pharmacotherapy. First-line psychotherapies include cognitive behavioral therapy (CBT) and interpersonal therapy (IPT). Antidepressant medications—SSRIs, SNRIs, atypical antidepressants, tricyclics, and MAO inhibitors—are chosen based on side-effect profiles, comorbidities, and prior response. For severe or treatment-resistant depression, augmentation strategies, electroconvulsive therapy (ECT), or newer neuromodulation techniques (TMS, vagus nerve stimulation) may be used. Safety planning and suicide risk assessment are critical. Long-term maintenance therapy reduces relapse risk; lifestyle measures, social support, and treatment of comorbid conditions are important.

Dementia (including Alzheimer’s and Vascular Dementia)

Symptoms Dementia is a syndrome of progressive cognitive decline affecting memory, language, executive function, and daily living activities. Alzheimer’s disease typically begins with short-term memory loss and progresses to impair multiple cognitive domains; vascular dementia often has a stepwise decline with focal neurological signs. Behavioral and psychiatric symptoms—apathy, agitation, psychosis, sleep disturbance—are common and distressing for caregivers. Functional decline leads to dependence and increased care needs.

Treatment Management includes symptomatic pharmacotherapy (cholinesterase inhibitors for mild-moderate Alzheimer’s; memantine for moderate-severe disease) and nonpharmacologic interventions (cognitive stimulation, structured routines, environmental modifications). Treating vascular risk factors (hypertension, diabetes, hyperlipidemia) can slow progression in vascular dementia. Behavioral symptoms are first addressed with nonpharmacologic strategies; antipsychotics are used cautiously for severe agitation due to increased mortality risk. Caregiver support, advanced care planning, and multidisciplinary coordination are essential.

Dermatitis (Eczema, Contact Dermatitis, Atopic Dermatitis)

Symptoms Dermatitis presents with erythema, pruritus (itching), scaling, vesicles, and sometimes oozing or crusting. Atopic dermatitis often begins in childhood with flexural involvement and intense itch; contact dermatitis results from exposure to irritants or allergens and follows a distribution corresponding to exposure. Chronic dermatitis leads to lichenification (thickened skin) and secondary infection.

Treatment Treatment focuses on trigger avoidance, skin hydration with emollients, and anti-inflammatory therapy. Topical corticosteroids are first-line for flares; topical calcineurin inhibitors are alternatives for sensitive areas. For severe or widespread disease, short courses of systemic corticosteroids or immunosuppressants (methotrexate, cyclosporine) may be used; newer biologics (dupilumab) are effective for moderate-severe atopic dermatitis. Antihistamines relieve itch; antibiotics treat

secondary bacterial infection. Patient education on gentle skin care and avoidance of irritants is key.

Diarrhea (Acute and Chronic)

Symptoms Diarrhea is increased stool frequency and/or liquidity. Acute infectious diarrhea often presents with abdominal cramps, fever, nausea, and sometimes bloody stools. Chronic diarrhea (lasting >4 weeks) may be due to inflammatory bowel disease, irritable bowel syndrome, malabsorption (celiac disease, pancreatic insufficiency), microscopic colitis, or medication side effects. Dehydration, electrolyte disturbances, and weight loss are important complications.

Treatment Acute infectious diarrhea is managed with rehydration (oral rehydration solutions), symptomatic care, and targeted antibiotics for specific bacterial pathogens or severe cases. Antimotility agents (loperamide) can be used in selected nonbloody cases. Chronic diarrhea requires evaluation to identify the cause (stool studies, blood tests, endoscopy, imaging) and disease-specific therapy: anti-inflammatories and immunosuppressants for IBD, gluten-free diet for celiac disease, pancreatic enzyme replacement for pancreatic insufficiency, bile acid sequestrants for bile acid diarrhea, and dietary modifications for IBS. Nutritional support and correction of electrolyte imbalances are essential.

Digestive Disorders (GERD, IBS, Peptic Ulcer Disease)

Symptoms Gastroesophageal reflux disease (GERD) causes heartburn, regurgitation, and sometimes chronic cough or laryngitis. Irritable bowel syndrome (IBS) presents with abdominal pain associated with altered bowel habits (constipation, diarrhea, or mixed) and bloating. Peptic ulcer disease causes epigastric pain, often related to meals, and may present with bleeding or perforation in complications.

Treatment GERD management includes lifestyle measures (weight loss, head-of-bed elevation, avoiding trigger foods) and acid suppression with proton pump inhibitors (PPIs) or H₂ blockers; refractory cases may need surgical fundoplication. IBS treatment is symptom-directed: dietary changes (low FODMAP), fiber, antispasmodics, laxatives or antidiarrheals, and psychological therapies (CBT). Peptic ulcer disease treatment targets *Helicobacter pylori* eradication when present (combination antibiotics plus PPI) and acid suppression; avoid NSAIDs or use protective strategies if NSAIDs are necessary.

Down Syndrome (Trisomy 21)

Symptoms Down syndrome is a genetic condition with characteristic facial features, intellectual disability of variable severity, hypotonia, and congenital anomalies (especially cardiac defects such as atrioventricular septal defects). Other issues

include thyroid dysfunction, hearing and vision problems, gastrointestinal malformations, and increased risk of leukemia and early-onset Alzheimer's disease.

Treatment Care is multidisciplinary and lifelong. Early intervention with developmental therapies (physical, occupational, speech) improves outcomes. Cardiac defects often require surgical correction in infancy. Regular screening for thyroid disease, hearing and vision problems, celiac disease, and hematologic abnormalities is recommended. Educational support, social services, and transition planning to adult care are important. Management focuses on maximizing independence and quality of life.

Drug Abuse (Prescription and Illicit Substances)

Symptoms Drug abuse manifests with behavioral changes (neglect of responsibilities, social withdrawal), physical signs (weight loss, track marks, nasal septum damage), cognitive impairment, mood instability, and tolerance/withdrawal phenomena. Specific substances produce characteristic effects: stimulants cause agitation and insomnia; opioids cause sedation and respiratory depression; benzodiazepines cause sedation and memory impairment.

Treatment Treatment includes medical stabilization, detoxification, pharmacotherapy for specific dependencies (buprenorphine/methadone for opioids, benzodiazepine tapering for sedative dependence), and psychosocial interventions (CBT, contingency management). Harm-reduction strategies (needle exchange, naloxone distribution) reduce morbidity and mortality. Long-term recovery involves relapse prevention, social support, vocational rehabilitation, and treatment of co-occurring mental health disorders.

Diphtheria (rare in vaccinated populations)

Symptoms Diphtheria, caused by *Corynebacterium diphtheriae*, presents with sore throat, low-grade fever, and a characteristic gray-white pseudomembrane on the tonsils and pharynx that can cause airway obstruction. Toxin-mediated complications include myocarditis and neuropathies. In countries with high vaccination coverage, diphtheria is rare but remains a risk in unvaccinated populations.

Treatment Immediate management includes administration of diphtheria antitoxin to neutralize circulating toxin and prompt antibiotic therapy (penicillin or erythromycin) to eradicate the organism and reduce transmission. Airway management may be necessary for obstruction. Close monitoring for cardiac and neurologic complications is required. Prevention relies on vaccination (DTaP/Tdap) and booster doses per schedule.

Section E — Medium entries (5 pages total; 4 topics)

Epilepsy

Symptoms & Treatment (concise) Epilepsy causes recurrent unprovoked seizures—transient disturbances of brain function manifesting as convulsions, staring spells, sensory phenomena, or loss of awareness. Diagnosis uses EEG and neuroimaging. Treatment centers on antiseizure medications (valproate, carbamazepine, levetiracetam, lamotrigine) tailored to seizure type; refractory cases may benefit from epilepsy surgery, vagus nerve stimulation, or ketogenic diet. Safety counseling and driving restrictions are important.

Eczema (Atopic Dermatitis)

Symptoms & Treatment (concise) Intensely itchy, inflamed skin with chronic relapsing course. Management: emollients, topical corticosteroids, topical calcineurin inhibitors, avoidance of triggers, and systemic therapies or biologics for severe disease.

Eating Disorders (Anorexia, Bulimia, Binge Eating)

Symptoms & Treatment (concise) Disordered eating with medical and psychiatric complications. Treatment is multidisciplinary: nutritional rehabilitation, psychotherapy (CBT, family-based therapy), medical monitoring, and pharmacotherapy for comorbid conditions.

Eye Diseases (Glaucoma, Macular Degeneration)

Symptoms & Treatment (concise) Glaucoma causes progressive peripheral vision loss; treated with topical pressure-lowering agents, laser, or surgery. Age-related macular degeneration causes central vision loss; management includes anti-VEGF injections for wet AMD and lifestyle/nutritional support for dry AMD.

Section F — Medium entries (5 pages total; 4 topics)

Flu (Influenza)

Symptoms & Treatment (concise) Abrupt fever, myalgias, cough, sore throat. Antiviral therapy (oseltamivir) for high-risk or severe cases; supportive care, vaccination for prevention.

Fibromyalgia

Symptoms & Treatment (concise) Widespread musculoskeletal pain, fatigue, sleep disturbance. Management: exercise, CBT, sleep hygiene, and medications (duloxetine, pregabalin) for symptom control.

Food Poisoning

Symptoms & Treatment (concise) Acute nausea, vomiting, diarrhea, abdominal pain after contaminated food. Rehydration, supportive care; antibiotics for specific bacterial pathogens when indicated.

Fractures (minor/major)

Symptoms & Treatment (concise) Acute pain, deformity, swelling. Immobilization, reduction, surgical fixation when necessary, and rehabilitation.

Section G — Medium entries (5 pages total; 4 topics)

Gallstones (Cholelithiasis)

Symptoms & Treatment (concise) Biliary colic—episodic RUQ pain after fatty meals; complications include cholecystitis, cholangitis, pancreatitis. Treatment: laparoscopic cholecystectomy for symptomatic disease; ERCP for choledocholithiasis.

Glaucoma (*covered above in E but expanded here if needed*)

Symptoms & Treatment (concise) See Section E.

Gout

Symptoms & Treatment (concise) Acute monoarticular arthritis (podagra), intense pain, swelling. Acute treatment: NSAIDs, colchicine, corticosteroids. Long-term: urate-lowering therapy (allopurinol, febuxostat) and lifestyle modification.

Gastroenteritis

Symptoms & Treatment (concise) Acute diarrhea, vomiting, abdominal cramps. Rehydration, supportive care; antibiotics for specific bacterial causes.

Section H — Full expanded content (20 pages; 10 topics)

Heart Disease (Ischemic Heart Disease, Coronary Artery Disease)

Symptoms Ischemic heart disease commonly presents with exertional chest pain (angina), shortness of breath, fatigue, and in acute coronary syndromes with severe chest pain, diaphoresis, nausea, and syncope. Atypical presentations occur in women, elderly, and diabetics. Chronic ischemia leads to heart failure and arrhythmias.

Treatment Management includes risk factor modification (smoking cessation, lipid control, blood pressure management, diabetes control), antiplatelet therapy (aspirin), statins, antianginal medications (beta-blockers, nitrates, calcium channel blockers), and revascularization (PCI or CABG) for significant coronary stenosis. Acute MI requires rapid reperfusion (PCI or thrombolysis), antithrombotic therapy, and intensive care. Cardiac rehabilitation improves outcomes.

Hepatitis (A, B, C, Autoimmune, Alcoholic)

Symptoms Viral hepatitis presents with fatigue, anorexia, nausea, right upper quadrant discomfort, jaundice, dark urine, and pale stools. Chronic hepatitis may be asymptomatic until advanced liver disease develops.

Treatment Acute viral hepatitis A is supportive; hepatitis B and C have effective antiviral therapies (direct-acting antivirals for HCV; nucleos(t)ide analogues for HBV) to reduce progression to cirrhosis. Vaccination prevents hepatitis A and B. Management of alcoholic hepatitis includes abstinence and supportive care; autoimmune hepatitis requires immunosuppression.

HIV/AIDS (*covered in A but expanded for completeness*)

Symptoms & Treatment (concise) See Section A for detailed management; ART, opportunistic infection prophylaxis, and monitoring are central.

Hypertension (*covered in B but expanded here*)

Symptoms & Treatment (concise) See Section B.

Hemorrhoids

Symptoms Rectal bleeding, pain, itching, and prolapse. External hemorrhoids cause perianal pain; internal hemorrhoids may bleed with defecation.

Treatment Conservative care: fiber, stool softeners, sitz baths, topical agents. Office procedures (rubber band ligation, sclerotherapy) for persistent symptomatic hemorrhoids; surgical hemorrhoidectomy for severe or recurrent disease.

Hernia (Inguinal, Femoral, Ventral)

Symptoms Bulge in the groin or abdominal wall, discomfort with straining; incarcerated hernia causes severe pain and obstruction.

Treatment Elective surgical repair (open or laparoscopic) for symptomatic hernias; urgent surgery for incarceration or strangulation.

Thyroid Disorders (Hypothyroidism, Hyperthyroidism)

Symptoms Hypothyroidism: fatigue, weight gain, cold intolerance, constipation, bradycardia. Hyperthyroidism: weight loss, heat intolerance, palpitations, tremor, anxiety.

Treatment Hypothyroidism: levothyroxine replacement with dose titration by TSH. Hyperthyroidism: antithyroid drugs (methimazole, propylthiouracil), radioactive iodine ablation, or surgery for definitive treatment; beta-blockers for symptomatic control.

Hair Loss (Alopecia Areata, Androgenetic Alopecia)

Symptoms Patchy hair loss (alopecia areata) or progressive thinning (androgenetic alopecia). Psychological impact can be significant.

Treatment Topical or intralesional corticosteroids for alopecia areata; minoxidil and finasteride for androgenetic alopecia; hair transplantation and cosmetic options for advanced cases.

Headaches (Tension, Migraine, Cluster)

Symptoms Tension headaches cause bilateral pressure; migraines cause unilateral throbbing pain with photophobia, phonophobia, and sometimes aura; cluster headaches are severe, unilateral, and associated with autonomic features.

Treatment Acute migraine: NSAIDs, triptans, antiemetics. Preventive therapy: beta-blockers, antiepileptics, CGRP inhibitors. Tension headaches: analgesics, stress management. Cluster headaches: oxygen therapy, triptans, and preventive verapamil.

Hearing Loss

Symptoms Gradual or sudden reduction in hearing, tinnitus, difficulty understanding speech. Causes include presbycusis, noise exposure, otitis media, ototoxic drugs.

Treatment Evaluation with audiometry; hearing aids, cochlear implants for severe sensorineural loss; medical or surgical treatment for conductive causes (e.g., otosclerosis, chronic otitis media).

Section I — (not in original plan but include common topics if needed)

(If you want Section I expanded, tell me and I'll add it; otherwise I'll continue to J–K as per your plan.)

Section M — Full expanded content (20 pages; 10 topics)

Migraine

Symptoms Recurrent moderate-to-severe headaches, often unilateral and pulsatile, associated with nausea, vomiting, photophobia, and phonophobia; some patients experience aura (visual or sensory disturbances) preceding headache.

Treatment Acute therapy: NSAIDs, triptans, antiemetics. Preventive therapy for frequent attacks: beta-blockers, antiepileptics (topiramate), antidepressants (amitriptyline), and CGRP monoclonal antibodies. Lifestyle triggers identification and management, sleep hygiene, and stress reduction are important.

Multiple Sclerosis (MS)

Symptoms Demyelinating disease causing sensory disturbances, weakness, visual loss (optic neuritis), ataxia, bladder dysfunction, and cognitive changes. Symptoms often remit and relapse in relapsing-remitting MS.

Treatment Acute relapses: high-dose corticosteroids. Disease-modifying therapies (interferons, glatiramer acetate, monoclonal antibodies, oral agents) reduce relapse

rate and slow progression. Symptomatic management (spasticity, fatigue, bladder dysfunction) and multidisciplinary rehabilitation are essential.

Malaria

Symptoms Fever, chills, sweats, headache, myalgias, and in severe cases anemia, jaundice, cerebral involvement, and multi-organ failure. Travel history to endemic areas is key.

Treatment Prompt diagnosis with blood smear or rapid tests. Uncomplicated malaria treated with artemisinin-based combination therapies (ACTs); severe malaria requires intravenous artesunate and supportive care. Prevention includes prophylaxis for travelers and vector control.

Mental Health Disorders (General overview)

Symptoms & Treatment (concise) Covers anxiety, depression, psychotic disorders, personality disorders. Treatment: psychotherapy, pharmacotherapy, social support, and crisis intervention when needed.

Measles

Symptoms & Treatment (concise) Fever, cough, coryza, conjunctivitis, Koplik spots, and maculopapular rash. Supportive care; vitamin A reduces morbidity. Vaccination prevents disease.

Meningitis

Symptoms & Treatment (concise) Fever, headache, neck stiffness, photophobia, altered mental status. Bacterial meningitis requires urgent empiric IV antibiotics and sometimes adjunctive corticosteroids; viral meningitis is usually supportive care.

Muscular Dystrophy

Symptoms & Treatment (concise) Progressive muscle weakness, gait disturbance, respiratory compromise in advanced disease. Management: multidisciplinary care, physiotherapy, respiratory support, and genetic counseling; disease-specific therapies emerging for some types.

Myopia (Nearsightedness)

Symptoms & Treatment (concise) Blurry distance vision; corrected with glasses, contact lenses, or refractive surgery; myopia control strategies (orthokeratology, low-dose atropine) for children.

Mouth Ulcers (Aphthous Stomatitis)

Symptoms & Treatment (concise) Painful oral ulcers; topical corticosteroids, antiseptic mouthwashes, and addressing nutritional deficiencies help; recurrent cases require evaluation.

Menopause

Symptoms & Treatment (concise) Vasomotor symptoms (hot flashes), sleep disturbance, mood changes, vaginal dryness, osteoporosis risk. Management: lifestyle measures, hormone replacement therapy (individualized), nonhormonal options for vasomotor symptoms, bone health strategies.

Section N — Medium entries (5 pages total; 4 topics)

Nutrition (Malnutrition, Obesity)

Symptoms & Treatment (concise) Malnutrition: weight loss, muscle wasting; treat with nutritional rehabilitation and address underlying causes. Obesity: excess weight with metabolic complications; treat with lifestyle, pharmacotherapy, and bariatric surgery when indicated.

Neuropathy (Peripheral Neuropathy)

Symptoms & Treatment (concise) Numbness, tingling, burning pain, weakness. Treat underlying cause (diabetes, toxins), symptomatic pain control (gabapentin, duloxetine), and rehabilitation.

Nasal Allergies (Allergic Rhinitis)

Symptoms & Treatment (concise) Sneezing, rhinorrhea, congestion; treat with intranasal steroids, antihistamines, and immunotherapy.

Nephritis (Glomerulonephritis)

Symptoms & Treatment (concise) Hematuria, proteinuria, edema, hypertension. Management depends on cause: immunosuppression for immune-mediated disease, blood pressure control, and renal support.

Section O — Medium entries (5 pages total; 4 topics)

Obesity

Symptoms & Treatment (concise) Excess adiposity with metabolic risk. Treatment: diet, exercise, behavioral therapy, pharmacotherapy (GLP-1 agonists), and bariatric surgery for eligible patients.

Osteoporosis

Symptoms & Treatment (concise) Often asymptomatic until fracture; fragility fractures of hip, spine. Treatment: calcium/vitamin D, bisphosphonates, denosumab, lifestyle measures, fall prevention.

Oral Health (Dental Caries, Periodontitis)

Symptoms & Treatment (concise) Tooth pain, bleeding gums; treat with dental care, hygiene, and management of systemic contributors.

Otitis Media

Symptoms & Treatment (concise) Ear pain, fever, hearing loss in children. Treat with analgesics, watchful waiting, or antibiotics per guidelines; tympanostomy tubes for recurrent disease.

Section P — Medium entries (5 pages total; 4 topics)

Parkinson's Disease

Symptoms & Treatment (concise) Bradykinesia, rigidity, resting tremor, postural instability. Treatment: levodopa/carbidopa, dopamine agonists, MAO-B inhibitors, deep brain stimulation for selected patients, and multidisciplinary rehabilitation.

Pneumonia

Symptoms & Treatment (concise) Fever, cough, sputum, dyspnea, chest pain. Diagnosis by chest X-ray; treat with appropriate antibiotics, oxygen, and supportive care; vaccination prevents many cases.

Psoriasis

Symptoms & Treatment (concise) Well-demarcated scaly plaques; topical therapies, phototherapy, systemic agents (methotrexate, biologics) for moderate-severe disease.

Prostate Cancer

Symptoms & Treatment (concise) Often asymptomatic early; urinary symptoms or metastatic bone pain in advanced disease. Management: active surveillance, surgery, radiation, androgen deprivation therapy, and systemic agents for advanced disease.

Section R — Medium entries (5 pages total; 4 topics)

Rheumatoid Arthritis

Symptoms & Treatment (concise) Symmetric inflammatory polyarthritis with morning stiffness. Treat with DMARDs (methotrexate), biologics, NSAIDs, and rehabilitation.

Respiratory Infections (Bronchitis, Pneumonia)

Symptoms & Treatment (concise) See Bronchitis and Pneumonia sections; supportive care and targeted antibiotics when indicated.

Rickets (Vitamin D Deficiency in Children)

Symptoms & Treatment (concise) Bone pain, deformities, delayed growth. Treat with vitamin D and calcium supplementation and address dietary causes.

Rosacea

Symptoms & Treatment (concise) Facial flushing, papules, telangiectasia. Treat with topical metronidazole, azelaic acid, oral antibiotics for inflammatory lesions, and laser for telangiectasia.

Section S — Full expanded content (20 pages; 10 topics)

Stroke (Ischemic and Hemorrhagic)

Symptoms Sudden focal neurological deficits: weakness or numbness on one side, speech disturbance, visual loss, dizziness, severe headache (hemorrhagic). Time-sensitive emergency—"time is brain."

Treatment Immediate assessment with CT/MRI. Ischemic stroke: reperfusion with IV thrombolysis (if within window) and endovascular thrombectomy for large vessel occlusion; secondary prevention with antiplatelets, anticoagulation for cardioembolic sources, risk factor control, and rehabilitation. Hemorrhagic stroke: blood pressure control, neurosurgical intervention for large bleeds, and reversal of anticoagulation when applicable.

Schizophrenia

Symptoms & Treatment (concise) Psychosis with hallucinations, delusions, disorganized thought, negative symptoms. Antipsychotic medications, psychosocial interventions, and long-term support are central.

Skin Cancer (Melanoma, Basal Cell, Squamous Cell)

Symptoms & Treatment (concise) New or changing skin lesions; biopsy for diagnosis. Surgical excision is primary treatment; advanced melanoma treated with immunotherapy and targeted agents.

Stress (Acute and Chronic)

Symptoms & Treatment (concise) Psychological and somatic symptoms (sleep disturbance, headaches, GI upset). Management: stress reduction techniques, CBT, lifestyle changes, and treatment of comorbid conditions.

Sinusitis

Symptoms & Treatment (concise) Facial pain/pressure, nasal congestion, purulent nasal discharge. Supportive care, intranasal steroids, and antibiotics for bacterial cases.

Stomach Ulcers (Peptic Ulcer Disease)

Symptoms & Treatment (concise) Epigastric pain, dyspepsia, possible bleeding. Treat *H. pylori* if present, PPI therapy, and avoid NSAIDs.

Sexually Transmitted Infections (STIs)

Symptoms & Treatment (concise) Variable presentations: discharge, ulcers, systemic symptoms. Diagnosis with testing; treat with pathogen-specific antibiotics or antivirals and partner notification.

Scoliosis

Symptoms & Treatment (concise) Lateral spinal curvature; mild cases observed, bracing for progressive curves in adolescents, surgery for severe deformity.

Sleep Disorders (Insomnia, Sleep Apnea)

Symptoms & Treatment (concise) Insomnia: difficulty initiating/maintaining sleep—CBT for insomnia, sleep hygiene, short-term pharmacotherapy. Obstructive sleep apnea: snoring, daytime sleepiness—CPAP, weight loss, and surgical options.

Seizures

Symptoms & Treatment (concise) See Epilepsy; acute seizure management includes airway protection, benzodiazepines for prolonged seizures, and evaluation for underlying causes.

Section T — Medium entries (5 pages total; 4 topics)

Tuberculosis (TB)

Symptoms & Treatment (concise) Chronic cough, weight loss, night sweats, hemoptysis. Diagnosis with sputum testing and imaging. Treatment: multi-drug antitubercular therapy for months; public health measures for contact tracing.

Thyroid Disorders (*covered earlier*)

Symptoms & Treatment (concise) See Section H.

Trauma (Head, Chest, Abdominal)

Symptoms & Treatment (concise) Acute injury management follows ATLS: airway, breathing, circulation, imaging, surgical intervention as needed, and rehabilitation.

Tonsillitis

Symptoms & Treatment (concise) Sore throat, fever, swollen tonsils. Supportive care; antibiotics for bacterial cases (strep throat); tonsillectomy for recurrent severe episodes.

Section U — Medium entries (5 pages total; 4 topics)

Ulcers (Peptic and Pressure Ulcers)

Symptoms & Treatment (concise) Peptic ulcers: see Section S. Pressure ulcers: local wound care, pressure offloading, nutrition, and surgical debridement for severe wounds.

Urinary Tract Infections (UTIs) (*covered earlier*)

Symptoms & Treatment (concise) See Section B/Bladder infections.

Urticaria (Hives)

Symptoms & Treatment (concise) Itchy wheals; treat with antihistamines, identify triggers, and consider short-term corticosteroids for severe flares.

Uremia (Advanced Kidney Failure)

Symptoms & Treatment (concise) Nausea, anorexia, pruritus, confusion, fluid overload. Management includes dialysis or transplantation and supportive care.

Section V — Medium entries (5 pages total; 4 topics)

Varicose Veins

Symptoms & Treatment (concise) Leg heaviness, visible dilated veins, swelling. Conservative care (compression stockings), sclerotherapy, endovenous ablation, or surgery for symptomatic disease.

Vitamin Deficiencies (D, B12, Iron)

Symptoms & Treatment (concise) Varied presentations: bone pain, anemia, neuropathy. Treat with supplementation and address underlying causes.

Viral Infections (General)

Symptoms & Treatment (concise) Range from mild URIs to severe systemic disease; supportive care, antivirals for specific viruses, and vaccination for prevention.

Vaccination (Preventive Care)

Symptoms & Treatment (concise) Not a disease but essential preventive measure; follow national immunization schedules for children and adults.

Section W — Medium entries (5 pages total; 4 topics)

Weight Management

Symptoms & Treatment (concise) Obesity management: diet, exercise, behavioral therapy, pharmacotherapy, and bariatric surgery for eligible patients.

Whooping Cough (Pertussis)

Symptoms & Treatment (concise) Paroxysmal cough, inspiratory whoop; macrolide antibiotics and supportive care; vaccination prevents disease.

Warts (Viral Verrucae)

Symptoms & Treatment (concise) Skin lesions caused by HPV; treat with cryotherapy, topical salicylic acid, or procedural removal.

Worm Infestations (Helminths)

Symptoms & Treatment (concise) GI symptoms, anemia in heavy infestations; treat with antiparasitic agents (albendazole, mebendazole, praziquantel) and sanitation measures.

Section Q — Small entry (3 pages)

Q Fever

Symptoms Q fever, caused by *Coxiella burnetii*, presents with fever, severe headache, myalgias, and sometimes atypical pneumonia or hepatitis. Chronic Q fever can cause endocarditis in predisposed individuals.

Treatment Acute Q fever is treated with doxycycline; chronic infection (endocarditis) requires prolonged combination therapy (doxycycline plus hydroxychloroquine) and specialist management. Prevention includes occupational precautions for at-risk workers and control of animal reservoirs.

Section X — Small entry (3 pages)

Xeroderma and X-linked Disorders

Xeroderma (Xeroderma Pigmentosum) Symptoms A rare genetic disorder of DNA repair causing extreme sensitivity to UV light, early freckling, pigmentary changes, and a high risk of skin cancers at a young age.

Treatment Strict UV protection, regular dermatologic surveillance, early excision of suspicious lesions, and genetic counseling.

X-linked Disorders (Overview) Symptoms & Treatment (concise) A group of genetic conditions transmitted via the X chromosome (e.g., hemophilia, Duchenne muscular dystrophy). Management is disease-specific: factor replacement for hemophilia,

multidisciplinary care and emerging gene therapies for muscular dystrophy, and genetic counseling for families.

Section Y — Small entry (3 pages)

Yeast Infections (Candidiasis)

Symptoms Vaginal candidiasis: itching, thick white discharge, dysuria. Oropharyngeal candidiasis: white plaques, soreness. Invasive candidiasis: fever, sepsis in immunocompromised patients.

Treatment Topical or oral antifungals (fluconazole, topical azoles) for mucocutaneous disease; systemic antifungals (echinocandins, fluconazole) for invasive disease. Address predisposing factors (diabetes, antibiotics, immunosuppression).

Yellow Fever

Symptoms & Treatment (concise) Acute viral hemorrhagic fever with fever, jaundice, bleeding in severe cases. Supportive care; live attenuated vaccine is highly effective for prevention.

Section Z — Small entry (3 pages)

Zika Virus

Symptoms Often mild febrile illness with rash, conjunctivitis, arthralgia; infection during pregnancy can cause congenital Zika syndrome (microcephaly, neurologic defects).

Treatment Supportive care; prevention focuses on mosquito control and travel advisories for pregnant women.

Zinc Deficiency

Symptoms & Treatment (concise) Growth retardation, impaired wound healing, immune dysfunction. Treat with zinc supplementation and dietary correction.

Abdominal Pain

Symptoms Abdominal pain ranges from mild, intermittent cramping to severe, constant pain and may be localized (e.g., right lower quadrant) or diffuse. Associated features that help identify cause include nausea, vomiting, anorexia, fever, bloating, constipation or diarrhea, hematemesis or melena, urinary symptoms, and referred pain to the shoulder or back. Physical exam findings such as guarding, rebound tenderness, rigidity, abdominal distension, absent bowel sounds, or a palpable mass suggest peritonitis, obstruction, or an intra-abdominal process. Red flags requiring urgent evaluation include hypotension, high fever, progressive peritoneal signs, persistent vomiting, or signs of shock. **Treatment** Management is directed at the underlying

diagnosis: urgent surgical consultation and appendectomy for appendicitis; bowel rest, nasogastric decompression, and often surgery for mechanical obstruction; IV fluids, analgesia, and antibiotics for intra-abdominal sepsis; proton pump inhibitors and H. pylori eradication for peptic ulcer disease; and supportive care (oral rehydration, antiemetics) for viral gastroenteritis. Initial ED care emphasizes hemodynamic stabilization, analgesia, targeted imaging (ultrasound, CT), laboratory testing (CBC, electrolytes, liver enzymes, amylase/lipase), and timely specialty referral when indicated.

Abscess Skin and Internal

Symptoms Superficial skin abscesses present as a painful, erythematous, warm, fluctuant nodule often with surrounding cellulitis; systemic symptoms (fever, malaise) may accompany larger or deeper infections. Internal abscesses (peritonsillar, intra-abdominal, hepatic, pelvic) produce localized pain, fever, leukocytosis, and organ-specific signs such as dysphagia, abdominal tenderness, or jaundice. Chronic or recurrent abscesses raise concern for underlying conditions (diabetes, immunosuppression, foreign body, fistula). **Treatment** Superficial abscesses require incision and drainage as the primary therapy, with wound care and culture when indicated; oral antibiotics are added for systemic signs, extensive cellulitis, immunocompromise, or difficult-to-drain locations. Deep or visceral abscesses usually need image-guided percutaneous drainage or surgical drainage plus targeted intravenous antibiotics guided by culture and sensitivity. Address predisposing factors (glycemic control, removal of foreign bodies) and provide tetanus prophylaxis when appropriate.

Addison's Disease Primary Adrenal Insufficiency

Symptoms Chronic adrenal insufficiency causes progressive fatigue, anorexia, weight loss, abdominal pain, salt craving, orthostatic dizziness, hyperpigmentation of sun-exposed skin and mucosa, hyponatremia, and hyperkalemia. Acute adrenal crisis presents with severe hypotension, vomiting, abdominal pain, fever, confusion, and can be life-threatening if not treated promptly. Non-specific symptoms often delay diagnosis until biochemical testing reveals low cortisol and elevated ACTH in primary disease. **Treatment** Lifelong replacement therapy with glucocorticoids (hydrocortisone or equivalent) and mineralocorticoid (fludrocortisone) is required for primary Addison's disease, with dose adjustments during stress or illness. Acute adrenal crisis requires immediate IV hydrocortisone, aggressive fluid resuscitation with isotonic saline, correction of electrolytes, and treatment of precipitating causes (infection, trauma). Patient education on sick-day dosing, emergency injectable steroids, and medical alert identification is essential to prevent recurrent crises.

Aneurysm Abdominal Aortic and Cerebral

Symptoms Many aneurysms are clinically silent until enlargement or rupture. Abdominal aortic aneurysm (AAA) may produce a pulsatile abdominal mass, back or abdominal pain, or a sense of fullness; rupture causes sudden severe abdominal or back pain, hypotension, and shock. Cerebral (berry) aneurysm rupture causes an abrupt, severe “thunderclap” headache, neck stiffness, photophobia, nausea, vomiting, focal neurologic deficits, and decreased consciousness. Unruptured cerebral aneurysms may cause cranial nerve palsies or seizures if large. **Treatment** Management depends on size, location, and rupture risk. Small, asymptomatic AAAs are monitored with serial imaging and aggressive risk-factor modification (blood-pressure control, smoking cessation). Elective repair (endovascular aneurysm repair or open surgery) is recommended when diameter exceeds guideline thresholds or growth is rapid. Ruptured aneurysms require emergency resuscitation and urgent surgical or endovascular repair. For cerebral aneurysms, neurosurgical clipping or endovascular coiling is used based on anatomy and patient factors; blood-pressure control and neurosurgical ICU care are critical after rupture.

Amyotrophic Lateral Sclerosis ALS

Symptoms ALS is a progressive motor neuron disease causing painless weakness, muscle wasting, fasciculations, spasticity, dysarthria, dysphagia, and eventual respiratory muscle involvement. Symptoms often begin focally (limb or bulbar) and spread, with preserved sensation and autonomic function. Respiratory decline leads to orthopnea, morning headaches, and recurrent pulmonary infections; cognitive or behavioral changes occur in a subset. **Treatment** There is no cure; care is multidisciplinary and focuses on slowing progression, preserving function, and symptom control. Disease-modifying drugs (riluzole, edaravone) may modestly slow decline. Interventions include respiratory support (noninvasive ventilation, cough assist), nutritional support (enteral feeding when needed), physical and occupational therapy to maintain mobility and safety, management of spasticity and secretions, and early palliative care planning. Genetic counseling is offered when familial forms are suspected.

Angina Pectoris Stable and Unstable

Symptoms Angina presents as chest discomfort, pressure, or tightness precipitated by exertion or emotional stress and relieved by rest or nitrates; radiation to jaw, neck, shoulder, or arm and associated dyspnea, diaphoresis, or nausea may occur. Unstable angina features new-onset, worsening, or rest angina and may herald myocardial infarction. Atypical symptoms are common in women, elderly, and diabetics.

Treatment Immediate management of acute ischemia includes oxygen if hypoxic, sublingual nitroglycerin, aspirin, and rapid triage. Long-term therapy focuses on antianginal medications (beta-blockers, nitrates, calcium channel blockers), antiplatelet therapy, high-intensity statin therapy, ACE inhibitors for comorbidities,

smoking cessation, diabetes control, and structured exercise. Revascularization with percutaneous coronary intervention or coronary artery bypass grafting is indicated for refractory symptoms or significant ischemia/anatomic disease.

Anaphylaxis Severe Systemic Allergy

Symptoms Anaphylaxis is a rapid, life-threatening allergic reaction characterized by sudden onset of urticaria, angioedema, bronchospasm with wheeze and respiratory distress, hypotension, gastrointestinal symptoms (vomiting, diarrhea), and possible loss of consciousness. Onset is typically within minutes to hours after exposure to an allergen (foods, drugs, insect stings). Biphasic reactions can occur hours after initial recovery. **Treatment** Immediate intramuscular epinephrine is the first-line therapy and should not be delayed; secure airway, provide high-flow oxygen, and give aggressive IV fluids for hypotension. Adjunctive treatments include antihistamines and corticosteroids to reduce prolonged or biphasic reactions, bronchodilators for bronchospasm, and observation for at least 4–24 hours depending on severity. Patients require education on avoidance, prescription of epinephrine auto-injectors, and referral to allergy specialists for testing and immunotherapy when appropriate.

Aortic Stenosis Valvular Obstruction

Symptoms Aortic stenosis causes exertional dyspnea, angina, and syncope; a harsh systolic ejection murmur radiating to the carotids and a diminished carotid upstroke are classic exam findings. Advanced disease leads to heart failure symptoms, left ventricular hypertrophy, and reduced exercise tolerance. Symptoms often progress insidiously over years. **Treatment** Medical therapy treats symptoms and comorbidities but does not reverse valve obstruction. Severe symptomatic aortic stenosis requires valve replacement—surgical aortic valve replacement (SAVR) or transcatheter aortic valve replacement (TAVR)—selected based on surgical risk, anatomy, and patient preference. Perioperative risk assessment, optimization of heart failure, and long-term follow-up are essential.

Arrhythmia Including Atrial Fibrillation

Symptoms Arrhythmias produce palpitations, lightheadedness, syncope, chest discomfort, dyspnea, or may be asymptomatic and detected incidentally. Atrial fibrillation causes irregularly irregular pulse and increases stroke risk; ventricular tachycardia or fibrillation can cause hemodynamic collapse. Underlying symptoms depend on rate, rhythm, and cardiac function. **Treatment** Management is arrhythmia-specific: for atrial fibrillation, acute rate control (beta-blockers, nondihydropyridine calcium channel blockers) or rhythm control (electrical cardioversion, antiarrhythmic drugs) is chosen based on presentation; anticoagulation is indicated according to stroke risk (CHA₂DS₂-VASc). Catheter ablation is an option for symptomatic, drug-refractory cases. Life-threatening ventricular arrhythmias require

immediate defibrillation and advanced cardiac life support; implantable cardioverter-defibrillators (ICDs) prevent sudden death in high-risk patients.

Arthritis Osteoarthritis and Rheumatoid Arthritis

Symptoms Osteoarthritis presents with joint pain worsened by use, morning stiffness that improves with activity, crepitus, and limited range of motion—commonly affecting knees, hips, hands, and spine. Rheumatoid arthritis is an inflammatory symmetric polyarthritis with prolonged morning stiffness, joint swelling, systemic fatigue, and potential extra-articular manifestations (lung, eye, vasculitis). Both cause progressive functional impairment if untreated. **Treatment** Osteoarthritis management emphasizes weight loss, exercise, physical therapy, analgesics (acetaminophen, NSAIDs), topical agents, intraarticular corticosteroids or hyaluronic acid for flares, and joint replacement for end-stage disease. Rheumatoid arthritis requires early initiation of disease-modifying antirheumatic drugs (DMARDs) such as methotrexate, with escalation to biologic agents (TNF inhibitors, IL-6 inhibitors, B-cell agents) for inadequate response; short courses of corticosteroids and NSAIDs control symptoms while DMARDs take effect. Rehabilitation and monitoring for comorbidities are important.

Asthma Chronic Reactive Airways Disease

Symptoms Asthma is characterized by episodic wheeze, cough (often nocturnal), chest tightness, and shortness of breath with variable airflow obstruction and bronchial hyperresponsiveness. Triggers include allergens, viral infections, exercise, cold air, and occupational exposures. Severity ranges from intermittent mild symptoms to life-threatening exacerbations with respiratory failure. **Treatment** Acute exacerbations are treated with inhaled short-acting beta-agonists, systemic corticosteroids, oxygen, and monitoring for respiratory compromise. Long-term control uses inhaled corticosteroids as first-line controller therapy, with addition of long-acting beta-agonists, leukotriene receptor antagonists, or biologic agents (anti-IgE, anti-IL-5/IL-4R) for severe phenotypes. Patient education, inhaler technique, trigger avoidance, and an individualized asthma action plan reduce morbidity.

Atopic Dermatitis Eczema

Symptoms Atopic dermatitis presents with chronic, intensely pruritic, relapsing eczematous lesions—erythematous, scaly, and often lichenified—commonly affecting face and extensor surfaces in infants and flexural areas in older children and adults. Scratching leads to excoriation, secondary infection, and sleep disturbance; severe disease impairs quality of life. **Treatment** Emollient therapy and skin barrier repair are foundational. Topical corticosteroids treat flares; topical calcineurin inhibitors are steroid-sparing options for sensitive areas. For moderate-to-severe disease, systemic immunomodulators (cyclosporine, methotrexate) or biologic therapy (dupilumab) may

be indicated. Identify and manage triggers, treat secondary infections, and provide patient education on long-term skin care.

Atherosclerosis and Peripheral Arterial Disease PAD

Symptoms Atherosclerosis is the pathologic process underlying coronary, cerebral, and peripheral arterial disease and may be clinically silent until ischemic events occur. PAD typically presents with exertional claudication—reproducible calf pain with walking relieved by rest—diminished distal pulses, cool extremities, and in advanced disease nonhealing ulcers or gangrene. **Treatment** Aggressive risk-factor modification (smoking cessation, statin therapy, blood-pressure and diabetes control) is essential. Supervised exercise programs improve walking distance; antiplatelet therapy reduces cardiovascular events. Pharmacologic agents such as cilostazol may relieve claudication. Revascularization (endovascular angioplasty/stenting or surgical bypass) is reserved for lifestyle-limiting symptoms or critical limb ischemia.

Autism Spectrum Disorder Neurodevelopmental Condition

Symptoms Autism spectrum disorder is defined by persistent deficits in social communication and interaction and restricted, repetitive patterns of behavior, interests, or activities. Symptoms appear in early childhood and vary widely in severity, often accompanied by language delay, intellectual disability, sensory sensitivities, and comorbid ADHD or anxiety. Functional impairment across social, educational, and occupational domains is common. **Treatment** Early, intensive behavioral interventions (applied behavior analysis, speech and occupational therapy) improve outcomes. Individualized educational programs, social skills training, and family support are central. Pharmacotherapy targets associated symptoms (irritability, hyperactivity, anxiety) rather than core social deficits. Lifelong supports and vocational planning help maximize independence.

Avascular Necrosis Osteonecrosis

Symptoms Avascular necrosis causes insidious joint pain (hip most commonly) that worsens with weight bearing and progresses to limited range of motion and joint collapse if untreated. Early disease may have minimal radiographic changes; MRI is sensitive for early detection. Risk factors include corticosteroid use, alcohol excess, trauma, and thrombophilic disorders. **Treatment** Management aims to preserve the joint and prevent collapse. Early measures include reduced weight bearing, bisphosphonates in select cases, and core decompression to relieve intraosseous pressure. Advanced disease with subchondral collapse often requires joint replacement (total hip arthroplasty). Address modifiable risk factors and consider referral to orthopedic specialists early.

Alopecia Hair Loss Disorders

Symptoms Alopecia presents as patterned hair thinning (androgenetic alopecia), patchy non-scarring hair loss (alopecia areata), or diffuse shedding (telogen effluvium). Onset, distribution, presence of scalp inflammation or scarring, and associated systemic symptoms guide diagnosis. Psychological distress and reduced quality of life are common. **Treatment** Androgenetic alopecia: topical minoxidil and oral finasteride (men) slow progression and promote regrowth; hair transplantation is an option for stable cases. Alopecia areata: intralesional corticosteroids for limited disease, topical immunotherapy or systemic immunomodulators for extensive disease. Telogen effluvium management focuses on identifying and treating triggers (illness, medications, nutritional deficiencies) and reassurance as hair regrowth often occurs.

Acid Reflux Disease GERD and Peptic Ulcer Disease

Symptoms GERD causes heartburn, regurgitation, chest discomfort, and extra-esophageal symptoms such as chronic cough, hoarseness, or laryngitis. Peptic ulcer disease presents with epigastric pain, dyspepsia, and potential complications of bleeding or perforation. Alarm features include weight loss, anemia, dysphagia, or GI bleeding. **Treatment** Lifestyle modifications (weight loss, head-of-bed elevation, avoid late meals and trigger foods) and proton pump inhibitors (PPIs) are first-line for GERD. H. pylori testing and eradication therapy are essential for peptic ulcer disease caused by the organism; stop NSAIDs when possible and use PPI therapy for ulcer healing. Surgical or endoscopic interventions are reserved for refractory disease or complications.

Acute Respiratory Distress Syndrome ARDS

Symptoms ARDS is an acute, diffuse inflammatory lung injury causing severe hypoxemia, bilateral pulmonary infiltrates, and respiratory failure not fully explained by cardiac failure. Common precipitants include sepsis, aspiration, severe pneumonia, trauma, and transfusion-related lung injury. Patients develop progressive dyspnea, tachypnea, and refractory hypoxemia despite supplemental oxygen. **Treatment** Management is supportive in the ICU with lung-protective mechanical ventilation (low tidal volumes, appropriate PEEP), conservative fluid strategy, treatment of the underlying cause (antibiotics for sepsis), prone positioning for severe hypoxemia, and consideration of extracorporeal membrane oxygenation (ECMO) in refractory cases. Prevent and treat complications such as ventilator-associated pneumonia and barotrauma; early rehabilitation is important for recovery.

Bell's Palsy Idiopathic Facial Nerve Palsy

Symptoms Bell's palsy presents with acute unilateral facial weakness affecting both the upper and lower face, often with sudden onset over hours to days. Patients may have decreased lacrimation, hyperacusis, altered taste on the anterior tongue, and difficulty closing the eye, which risks corneal exposure. Most cases are idiopathic but

may follow viral illness. **Treatment** Early short-course oral corticosteroids improve the likelihood of full recovery; antivirals may be considered in severe cases though benefit is uncertain. Protect the cornea with lubricating drops, ointment, and eye patching as needed. Physical therapy and facial retraining may aid recovery; most patients recover partially or fully within weeks to months, though some have residual weakness.

Appendicitis Acute Appendiceal Inflammation

Symptoms Appendicitis typically begins with periumbilical pain that migrates to the right lower quadrant (McBurney point), accompanied by anorexia, nausea, vomiting, and low-grade fever. Physical exam may show localized tenderness, guarding, rebound, and positive signs such as Rovsing or psoas sign. Atypical presentations occur in children, elderly, and pregnant patients. **Treatment** Standard treatment is appendectomy (laparoscopic or open) for uncomplicated appendicitis; selected patients with early, uncomplicated disease may be managed nonoperatively with antibiotics and close follow-up. Perforated appendicitis with abscess often requires drainage and delayed appendectomy. Prompt diagnosis with clinical assessment and imaging (ultrasound or CT) reduces complications.

Bell's Palsy (Idiopathic Facial Nerve Palsy) Symptoms Bell's palsy presents with sudden onset unilateral facial weakness that affects both the upper and lower face, typically developing over hours to a day. Patients report inability to close the eye, drooping of the mouth, difficulty smiling, and impaired forehead movement on the affected side. Associated features can include decreased lacrimation, hyperacusis (sensitivity to sound), altered taste on the anterior two-thirds of the tongue, and mild pain or discomfort around the ear preceding weakness. On exam there is flaccid paralysis of facial muscles with preserved sensation; absence of forehead sparing helps distinguish peripheral facial nerve palsy from central lesions. Most cases are idiopathic but may follow viral prodromes (herpes simplex or zoster) or occur with diabetes or pregnancy. **Treatment** Early treatment with a short course of oral corticosteroids (prednisone) started within 72 hours of onset improves the likelihood of full recovery. Antiviral therapy (acyclovir or valacyclovir) may be added when herpes zoster or severe palsy is suspected, though benefit is modest. Eye care is essential: lubricating drops, ointment at night, and protective patching to prevent corneal drying and injury when eyelid closure is incomplete. Physical therapy and facial retraining exercises can aid functional recovery. Most patients recover substantially within weeks to months; referral to neurology or ENT is indicated for atypical features, progressive weakness, or lack of improvement.

Benign Prostatic Hyperplasia (BPH) Symptoms BPH is a nonmalignant enlargement of the prostate common in older men that produces lower urinary tract symptoms (LUTS). Typical complaints include urinary frequency, nocturia, urgency, weak urinary stream, hesitancy, straining, incomplete bladder emptying, and post-void dribbling.

Severe obstruction can lead to urinary retention, recurrent urinary tract infections, bladder stones, or renal impairment. Symptom severity is often graded using validated questionnaires (e.g., IPSS). **Treatment** Management is tailored to symptom severity and patient preference. Watchful waiting with lifestyle measures (fluid timing, reducing caffeine/alcohol) is appropriate for mild symptoms. Medical therapy includes alpha-1 blockers (tamsulosin, terazosin) for rapid symptom relief and 5-alpha-reductase inhibitors (finasteride, dutasteride) to reduce prostate volume over months and lower long-term risk of progression. Combination therapy is used for moderate-to-severe symptoms. Minimally invasive procedures (transurethral microwave therapy, UroLift) and surgical options (transurethral resection of the prostate, open prostatectomy) are reserved for refractory symptoms, complications, or very large prostates. Evaluate for prostate cancer when indicated by PSA, exam, or atypical features.

Birth Defects (Congenital Anomalies) Symptoms Birth defects encompass structural or functional anomalies present at birth that range from minor cosmetic differences to life-threatening malformations. Examples include neural tube defects (spina bifida), congenital heart defects (ventricular septal defect, tetralogy of Fallot), limb anomalies, cleft lip/palate, and chromosomal syndromes (Down syndrome). Clinical presentation varies by defect: cyanosis or heart murmur in cardiac lesions, neurologic deficits in neural tube defects, feeding difficulties with cleft palate, and growth or developmental delays with genetic syndromes. Many anomalies are detected prenatally by ultrasound or newborn screening. **Treatment** Management depends on the specific anomaly and severity. Some defects require immediate neonatal stabilization and surgical correction (complex congenital heart disease, intestinal atresia), while others benefit from staged surgical repair (cleft lip/palate) and multidisciplinary rehabilitation (physical therapy, speech therapy). Genetic counseling, early intervention services, and long-term follow-up for developmental, educational, and medical needs are essential. Prevention strategies include periconceptional folic acid to reduce neural tube defects, avoidance of teratogens, and management of maternal conditions (diabetes, infections).

Bleeding Disorders and Hemophilia Symptoms Bleeding disorders include inherited conditions (hemophilia A/B, von Willebrand disease) and acquired coagulopathies. Hemophilia typically presents with prolonged bleeding after trauma or surgery, spontaneous joint and muscle hemorrhages, and easy bruising; severity correlates with factor deficiency level. Von Willebrand disease causes mucocutaneous bleeding—epistaxis, menorrhagia, easy bruising—and may be mild and intermittent. Acquired disorders (liver disease, vitamin K deficiency, anticoagulant therapy) produce variable bleeding risk including GI bleeding and intracranial hemorrhage. **Treatment** Treatment is diagnosis-specific. Hemophilia A/B is treated with replacement of deficient clotting factor concentrates (factor VIII or IX), prophylactic regimens to prevent joint damage in severe disease, and desmopressin (DDAVP) for mild hemophilia A. Von Willebrand disease management includes DDAVP for responsive types, von Willebrand factor

concentrates for severe cases, and antifibrinolytic agents (tranexamic acid) for mucosal bleeding. Acquired coagulopathies require correction of underlying causes, vitamin K, plasma or prothrombin complex concentrates for urgent reversal, and careful management of anticoagulant therapy.

Sickle Cell Disease and Thalassemia (Inherited Hemoglobinopathies) Symptoms

Sickle cell disease causes episodic vaso-occlusive pain crises, chronic hemolytic anemia, fatigue, jaundice, splenic dysfunction with infection risk, acute chest syndrome, stroke, and end-organ damage. Thalassemias range from asymptomatic trait to transfusion-dependent major forms; severe disease presents with profound anemia, growth retardation, bone deformities from marrow expansion, and iron overload from transfusions. Both disorders often manifest in childhood. **Treatment** Sickle cell management includes pain control, hydration, oxygen, infection prevention (vaccination, penicillin prophylaxis in young children), hydroxyurea to reduce vaso-occlusive events, chronic transfusion for stroke prevention, and curative hematopoietic stem cell transplantation in selected patients. Thalassemia major requires regular transfusions with iron chelation therapy to prevent iron overload; splenectomy and bone marrow transplantation are options in selected cases. Newer gene-based therapies are emerging for both conditions.

Bone Marrow Transplantation (Hematopoietic Stem Cell Transplant) Symptoms

Indications for bone marrow transplantation include hematologic malignancies (leukemia, lymphoma), severe aplastic anemia, certain inherited metabolic or immune disorders, and some hemoglobinopathies. Patients considered for transplant may present with cytopenias, recurrent infections, bleeding, or disease-specific symptoms. Pretransplant evaluation assesses disease status, organ function, donor availability, and psychosocial readiness. **Treatment** Hematopoietic stem cell transplantation involves conditioning chemotherapy ± radiation to eradicate disease and immunosuppress the host, followed by infusion of donor or autologous stem cells. Posttransplant care focuses on infection prophylaxis, graft-versus-host disease prevention and management, monitoring for engraftment and complications (veno-occlusive disease, organ toxicity), and long-term surveillance for relapse and late effects. Outcomes depend on disease, donor match, patient age, and comorbidities.

Botulism (Clostridium botulinum Neurotoxin Poisoning) Symptoms

Botulism presents with acute, symmetric, descending flaccid paralysis beginning with cranial nerve dysfunction—diplopia, blurred vision, ptosis, dysarthria, dysphagia—followed by progressive limb weakness and respiratory muscle involvement that can lead to respiratory failure. Autonomic symptoms (dry mouth, constipation) are common. Onset varies by form: foodborne, wound, or infant botulism. Mental status remains intact.

Treatment Botulism is a medical emergency requiring prompt supportive care, including airway protection and mechanical ventilation when respiratory compromise

occurs. Administration of antitoxin (equine-derived or human botulism immune globulin for infant botulism) neutralizes circulating toxin but does not reverse established paralysis. Wound debridement and antibiotics are indicated for wound botulism (avoid aminoglycosides). Recovery may take weeks to months with intensive rehabilitation.

Bronchiectasis Symptoms Bronchiectasis is characterized by chronic productive cough with large volumes of purulent sputum, recurrent respiratory infections, hemoptysis, dyspnea, and fatigue. Physical exam may reveal crackles and wheeze; clubbing occurs in some patients. Causes include prior severe infections, cystic fibrosis, immune deficiency, or airway obstruction. CT chest demonstrates bronchial dilation and wall thickening. **Treatment** Management aims to control infection, reduce sputum burden, and prevent exacerbations. Airway clearance techniques (physiotherapy, postural drainage), regular exercise, and mucolytics help sputum clearance. Antibiotics treat acute exacerbations and may be used long-term in selected patients with frequent infections; inhaled antibiotics are an option for chronic *Pseudomonas* infection. Vaccination, bronchodilators for coexisting airway disease, and surgical resection for localized disease refractory to medical therapy are additional strategies.

Bronchiolitis (Infants and Young Children) Symptoms Bronchiolitis is an acute viral lower respiratory tract infection most commonly caused by respiratory syncytial virus (RSV) in infants and young children. It presents with rhinorrhea, cough, tachypnea, wheeze, crackles, feeding difficulty, and varying degrees of respiratory distress. High-risk infants (premature, chronic lung disease, congenital heart disease) are more likely to require hospitalization. **Treatment** Care is primarily supportive: hydration, nasal suctioning, oxygen for hypoxemia, and monitoring for respiratory failure. Bronchodilators and corticosteroids are not routinely recommended but may be trialed in selected patients with reactive airways. Severe cases require hospitalization for respiratory support, including high-flow nasal cannula or mechanical ventilation. Prevention includes hand hygiene and RSV prophylaxis (palivizumab) for high-risk infants.

Bacterial Meningitis (Acute Meningitis) Symptoms Acute bacterial meningitis presents with fever, severe headache, neck stiffness (nuchal rigidity), photophobia, altered mental status, and sometimes focal neurologic deficits or seizures. Infants may show nonspecific signs—poor feeding, irritability, bulging fontanelle. Rapid progression and high morbidity/mortality make early recognition critical. Common pathogens vary by age (Group B *Streptococcus* and *E. coli* in neonates; *Streptococcus pneumoniae* and *Neisseria meningitidis* in older children and adults). **Treatment** Immediate empiric intravenous antibiotics that penetrate the CSF are essential after obtaining blood cultures and, when safe, lumbar puncture. Empiric regimens are age- and risk-based and often include a third-generation cephalosporin plus vancomycin; ampicillin is

added for *Listeria* coverage in older adults and neonates. Adjunctive dexamethasone reduces neurologic complications in pneumococcal meningitis when given before or with the first antibiotic dose. Supportive care, close monitoring for complications (seizures, increased intracranial pressure), and public health measures (prophylaxis for close contacts in meningococcal disease) are required.

Bacterial Endocarditis (Infective Endocarditis) Symptoms Infective endocarditis presents with fever, new or changing heart murmur, fatigue, anorexia, weight loss, and signs of embolic phenomena (splinter hemorrhages, Janeway lesions, stroke, splenic infarcts). Subacute presentations may be indolent, while acute staphylococcal infections can be fulminant. Risk factors include prosthetic valves, prior endocarditis, intravenous drug use, and structural heart disease. Blood cultures and echocardiography (transesophageal preferred) are key to diagnosis. **Treatment** Management requires prolonged, targeted intravenous antibiotics based on culture and sensitivity, often for 4–6 weeks. Early involvement of infectious disease and cardiothoracic surgery teams is important; surgical valve repair or replacement is indicated for heart failure, uncontrolled infection, large vegetations with embolic risk, or prosthetic valve involvement. Supportive care and prevention (dental hygiene, appropriate prophylaxis in high-risk procedures) are part of comprehensive management.

Bladder Cancer Symptoms Bladder cancer most commonly presents with painless gross hematuria. Other symptoms include urinary frequency, urgency, dysuria, and pelvic pain in advanced disease. Risk factors include tobacco smoking, occupational exposures (aromatic amines), chronic bladder inflammation, and prior pelvic radiation. Most tumors are urothelial (transitional cell) carcinomas. **Treatment** Evaluation includes cystoscopy with biopsy and imaging to stage disease. Non-muscle-invasive tumors are treated with transurethral resection and intravesical therapy (BCG or chemotherapy) to reduce recurrence. Muscle-invasive disease often requires radical cystectomy with urinary diversion, systemic chemotherapy for locally advanced or metastatic disease, and bladder-sparing multimodality therapy in selected patients. Surveillance for recurrence is intensive due to high recurrence rates.

Brain Tumors (Primary and Metastatic) Symptoms Symptoms depend on tumor location, size, and growth rate and may include persistent headache (often worse in the morning), seizures, focal neurologic deficits (weakness, sensory loss, aphasia, visual field deficits), cognitive or personality changes, nausea, and signs of increased intracranial pressure. Primary brain tumors (gliomas, meningiomas) and metastatic lesions from lung, breast, melanoma, kidney, or colon are common. Imaging with MRI is the diagnostic modality of choice. **Treatment** Management is multidisciplinary and may include surgical resection for diagnosis and cytoreduction, radiation therapy (conformal, stereotactic radiosurgery), and systemic therapies (chemotherapy, targeted

agents, immunotherapy) depending on tumor histology and molecular markers. Supportive care addresses seizures (antiepileptics), corticosteroids for vasogenic edema, rehabilitation, and palliative measures. Prognosis varies widely by tumor type, grade, and patient factors.

❓ **Cancer (General) Symptoms** Cancer symptoms vary by organ and tumor type but commonly include unexplained weight loss, persistent fatigue, localized pain or mass, changes in bowel or bladder habits, persistent cough or hoarseness, unexplained bleeding, and systemic signs such as fever or night sweats. Early disease may be asymptomatic; screening tests (mammography, colonoscopy, Pap test) detect many cancers before symptoms appear. **Treatment** Treatment is tumor-specific and may combine surgery, radiation therapy, systemic chemotherapy, targeted biologic agents, immunotherapy, and supportive/palliative care. Staging guides therapy; multidisciplinary teams tailor curative or palliative approaches, manage treatment toxicities, and provide survivorship planning.

❓ **Cataracts Symptoms** Progressive, painless blurring of vision, glare (especially at night), halos around lights, faded colors, and difficulty reading or driving. Symptoms worsen gradually as the lens becomes opacified. **Treatment** Early cataracts may be managed with refractive correction and brighter lighting; definitive treatment is surgical removal of the cloudy lens with intraocular lens implantation (phacoemulsification), which restores vision in the majority of patients. Perioperative care includes topical antibiotics and anti-inflammatory agents.

❓ **Cellulitis Symptoms** Acute, spreading erythema, warmth, swelling, and tenderness of skin and subcutaneous tissues; fever and regional lymphadenopathy may accompany more severe infection. Commonly follows a break in the skin. **Treatment** Empiric oral or IV antibiotics targeting streptococci and staphylococci (choice guided by severity and MRSA prevalence), elevation and immobilization of the limb, analgesia, and drainage if an abscess is present. Hospitalization for systemic signs or immunocompromise.

❓ **Cerebral Aneurysm (Subarachnoid Hemorrhage) Symptoms** Unruptured aneurysms are often asymptomatic; rupture causes sudden, severe “thunderclap” headache, neck stiffness, photophobia, nausea, vomiting, loss of consciousness, and focal neurologic deficits. Early recognition is critical. **Treatment** Emergency stabilization, neuroimaging (CT, angiography), neurosurgical or endovascular repair (clipping or coiling) to secure the aneurysm, and intensive care management to control intracranial pressure, vasospasm, and complications. Long-term follow-up and risk-factor control are essential.

❓ **Cerebral Palsy Symptoms** Nonprogressive motor impairment from early brain injury presenting with spasticity, dyskinesia, or ataxia; delayed motor milestones, abnormal

tone, and gait disturbances; associated impairments may include intellectual disability, seizures, vision or hearing problems, and speech difficulties. **Treatment** Multidisciplinary rehabilitation (physical, occupational, speech therapy), spasticity management (oral medications, botulinum toxin, intrathecal baclofen), orthopedic interventions for contractures, seizure control, and educational supports to maximize function and participation.

❓ **Cervical Cancer Symptoms** Early cervical cancer is often asymptomatic; later signs include abnormal vaginal bleeding (postcoital or intermenstrual), pelvic pain, and abnormal Pap test results. Screening with Pap smears and HPV testing reduces incidence and mortality. **Treatment** Management depends on stage: local excision (LEEP, conization) for preinvasive disease; radical surgery (hysterectomy) and/or radiation with concurrent chemotherapy for invasive disease; fertility-sparing options considered in selected early cases. Follow-up includes surveillance for recurrence.

❓ **Chickenpox (Varicella) and Shingles (Herpes Zoster) Symptoms** Varicella: fever and a pruritic vesicular rash in successive crops; complications include bacterial superinfection and pneumonia in adults. Herpes zoster: painful, unilateral dermatomal vesicular eruption with risk of postherpetic neuralgia. **Treatment** Varicella is usually supportive (antihistamines, skin care); antivirals (acyclovir) for adults or immunocompromised patients. Herpes zoster: early antiviral therapy reduces severity and duration; analgesics, topical care, and neuropathic pain agents (gabapentin, TCAs) for postherpetic neuralgia. Vaccination prevents or reduces severity.

❓ **Chronic Kidney Disease (Chronic Kidney Failure) Symptoms** Gradual fatigue, anorexia, nausea, pruritus, edema, anemia, electrolyte disturbances, and uremic symptoms as glomerular filtration declines; hypertension and bone-mineral disorders are common. **Treatment** Slow progression with blood-pressure control (ACE inhibitors/ARBs), glycemic control in diabetes, dietary measures, anemia management (erythropoiesis-stimulating agents, iron), and timely referral for renal replacement therapy (dialysis or transplantation) when indicated. Address reversible causes and provide cardiovascular risk reduction.

❓ **Chronic Obstructive Pulmonary Disease (COPD) / Chronic Bronchitis / Emphysema Symptoms** Chronic productive cough, progressive dyspnea, wheeze, and exercise intolerance; exacerbations with increased sputum purulence and breathlessness; advanced disease leads to hypoxemia and hypercapnia. **Treatment** Smoking cessation is paramount. Pharmacotherapy includes bronchodilators (short- and long-acting beta-agonists, anticholinergics), inhaled corticosteroids for frequent exacerbations, pulmonary rehabilitation, oxygen therapy for chronic hypoxemia, and vaccinations. Manage exacerbations with bronchodilators, systemic steroids, and antibiotics when indicated.

❓ **Cirrhosis and Chronic Liver Disease Symptoms** Fatigue, anorexia, weight loss, jaundice, pruritus, abdominal distension from ascites, easy bruising, and encephalopathy in advanced disease; portal hypertension causes variceal bleeding and splenomegaly. **Treatment** Treat underlying cause (viral hepatitis therapy, alcohol cessation, metabolic control). Manage complications: diuretics and paracentesis for ascites, endoscopic therapy and beta-blockers for varices, lactulose for encephalopathy, and liver transplantation for decompensated cirrhosis. Preventive measures include vaccination and alcohol counseling.

❓ **Colon Cancer (Colorectal Cancer) Symptoms** Painless rectal bleeding, change in bowel habits, unexplained iron-deficiency anemia, abdominal pain, and weight loss; many early cancers are detected by screening colonoscopy or fecal occult blood testing. **Treatment** Surgical resection is the mainstay for localized disease; adjuvant chemotherapy for stage II/III disease and systemic therapy for metastatic disease (chemotherapy, targeted agents, immunotherapy depending on molecular profile). Screening and polypectomy prevent many cancers.

❓ **Concussion and Traumatic Brain Injury (TBI) Symptoms** Transient confusion, headache, dizziness, nausea, memory impairment, sleep disturbance, and sometimes loss of consciousness after head trauma; persistent post-concussive symptoms can last weeks to months. Severe TBI causes focal deficits, prolonged unconsciousness, and cognitive/behavioral changes. **Treatment** Initial assessment for red flags and imaging when indicated; cognitive and physical rest followed by graded return to activity for concussion. Severe TBI requires neurosurgical and intensive care management, rehabilitation (physical, occupational, speech therapy), and long-term neuropsychological support.

❓ **Congenital Heart Disease Symptoms** Presentations range from cyanosis and heart failure in neonates to exertional dyspnea, murmurs, and exercise intolerance in older children and adults; specific defects (VSD, ASD, tetralogy of Fallot, transposition) have characteristic clinical patterns. **Treatment** Management includes medical therapy for heart failure and arrhythmias, catheter-based interventions (device closure, balloon valvuloplasty), and surgical repair or palliation depending on defect type and severity. Lifelong cardiology follow-up is often required.

❓ **Coronary Artery Disease (Ischemic Heart Disease) Symptoms** Stable angina (exertional chest discomfort), acute coronary syndromes (unstable angina, myocardial infarction) with chest pain, diaphoresis, dyspnea, nausea, and sometimes atypical presentations in women and elderly. Chronic ischemia causes heart failure and arrhythmias. **Treatment** Acute MI requires rapid reperfusion (PCI or thrombolysis), antiplatelet therapy, anticoagulation, and supportive care. Long-term management includes antiplatelet agents, statins, beta-blockers, ACE inhibitors, lifestyle

modification, cardiac rehabilitation, and revascularization when indicated. Risk-factor control is central to prevention.

❑ **Crohn's Disease (Inflammatory Bowel Disease) Symptoms** Chronic abdominal pain, diarrhea (often bloody), weight loss, fever, and perianal disease; transmural inflammation can cause strictures, fistulae, and malabsorption. Extraintestinal manifestations include arthritis, skin lesions, and eye inflammation. **Treatment** Induction with corticosteroids for flares; maintenance with immunomodulators (azathioprine, methotrexate) or biologic agents (anti-TNF, anti-integrin, anti-IL-12/23) to control inflammation and prevent complications. Nutritional support, antibiotics for abscesses, and surgery for strictures or refractory disease are part of care.

❑ **Cystic Fibrosis Symptoms** Chronic productive cough, recurrent pulmonary infections, bronchiectasis, malabsorption with steatorrhea and failure to thrive, male infertility, and salt-loss in infancy. Progressive lung disease is the major cause of morbidity and mortality. **Treatment** Multidisciplinary care: airway clearance techniques, inhaled bronchodilators and mucolytics, chronic suppressive antibiotics for *Pseudomonas*, pancreatic enzyme replacement, nutritional support, CFTR-modulator therapies for eligible mutations, and lung transplantation for end-stage disease. Vaccination and infection control are important.

❑ **Deep Vein Thrombosis (DVT) and Pulmonary Embolism (PE) Symptoms** DVT: unilateral leg swelling, pain, warmth, and erythema. PE: sudden dyspnea, pleuritic chest pain, tachycardia, hypoxemia, and sometimes syncope. Both can be life-threatening. **Treatment** Anticoagulation (heparin followed by oral anticoagulants or direct oral anticoagulants) is the mainstay; thrombolysis for massive PE with hemodynamic compromise; IVC filters for contraindications to anticoagulation; compression stockings and risk-factor modification for prevention. Evaluate for provoking factors and consider extended therapy when indicated.

❑ **Dementia (including Alzheimer Disease) Symptoms** Progressive decline in memory, executive function, language, and visuospatial skills with behavioral changes; Alzheimer disease is the most common cause, but vascular, Lewy body, and frontotemporal dementias have distinct features. Functional impairment in daily activities is typical. **Treatment** Cholinesterase inhibitors and memantine provide modest symptomatic benefit in Alzheimer disease; address reversible causes, manage behavioral symptoms with nonpharmacologic strategies first, and provide caregiver support, safety planning, and long-term care coordination.

❑ **Depressive Disorders (Major Depression) Symptoms** Persistent low mood, anhedonia, sleep and appetite changes, fatigue, impaired concentration, feelings of worthlessness or guilt, and suicidal ideation. Severity ranges from mild to severe with psychotic features. **Treatment** Evidence-based care includes psychotherapy (CBT,

interpersonal therapy), antidepressant medications (SSRIs, SNRIs, others), and for severe or treatment-resistant cases, electroconvulsive therapy or newer neuromodulation techniques. Safety assessment and management of comorbidities are essential.

🔍 **Diabetes Mellitus (Type 1 and Type 2) Symptoms** Polyuria, polydipsia, polyphagia, weight loss (type 1), fatigue, blurred vision, and complications such as neuropathy, nephropathy, retinopathy, and cardiovascular disease with chronic hyperglycemia.

Treatment Type 1 requires lifelong insulin therapy, glucose monitoring, and education. Type 2 management includes lifestyle modification, oral agents (metformin first-line), GLP-1 receptor agonists or SGLT2 inhibitors for cardiovascular/renal benefit, and insulin when needed. Preventive care for complications and individualized glycemic targets are important.

🔍 **Diverticulitis (Diverticulosis and Complications) Symptoms** Diverticulosis is often asymptomatic; diverticulitis presents with left lower-quadrant abdominal pain, fever, leukocytosis, and possible complications—abscess, perforation, fistula, or obstruction.

Treatment Uncomplicated diverticulitis: bowel rest, antibiotics, and outpatient management for mild cases. Complicated disease may require percutaneous drainage of abscesses or surgical resection for perforation, recurrent episodes, or fistula. Long-term prevention includes high-fiber diet and risk-factor modification.

🔍 **Epilepsy and Seizure Disorders Symptoms** Recurrent unprovoked seizures with variable manifestations—focal motor or sensory events, impaired awareness, generalized tonic-clonic convulsions. Postictal confusion and injury risk are common. Etiologies include structural brain lesions, genetic syndromes, metabolic disturbances, and idiopathic causes. **Treatment** Antiepileptic drugs chosen by seizure type (carbamazepine, valproate, levetiracetam, etc.), lifestyle counseling, seizure precautions, and for refractory focal epilepsy, surgical resection, vagus nerve stimulation, or ketogenic diet in selected patients. Address reversible triggers and comorbidities.

🔍 **Fibromyalgia Symptoms** Chronic widespread musculoskeletal pain, fatigue, sleep disturbance, cognitive complaints (“fibro fog”), and multiple tender points; symptoms often follow physical or emotional stressors. **Treatment** Multimodal approach: patient education, graded exercise, cognitive behavioral therapy, sleep hygiene, and pharmacotherapy (duloxetine, pregabalin, low-dose amitriptyline) for symptom control. Address comorbid mood disorders and encourage self-management strategies.

🔍 **Gallstones (Cholelithiasis) and Cholecystitis Symptoms** Biliary colic—episodic right upper-quadrant or epigastric pain often after fatty meals; acute cholecystitis causes persistent severe RUQ pain, fever, leukocytosis, and Murphy sign. Complications include choledocholithiasis and pancreatitis. **Treatment** Symptomatic

gallstones: elective laparoscopic cholecystectomy. Acute cholecystitis: initial resuscitation, antibiotics, and early cholecystectomy or percutaneous cholecystostomy in high-risk patients. ERCP for common bile duct stones when present.

Headache (Primary and Secondary Headaches) Symptoms Headache presents as pain in the head or upper neck and ranges from episodic tension-type headaches (bilateral, pressing/tightening) to migraine (unilateral, pulsatile, moderate-to-severe, often with photophobia, phonophobia, nausea, and aura) and cluster headaches (severe unilateral periorbital pain with autonomic features). Secondary headaches arise from underlying causes—sinusitis, meningitis, intracranial hemorrhage, tumor, temporal arteritis, or medication overuse—and may have red flags: sudden onset “thunderclap” pain, focal neurologic signs, fever, progressive worsening, or new headache in older adults. **Treatment** Primary headaches: acute migraine therapy includes NSAIDs, triptans, antiemetics, and rest in a dark, quiet environment; preventive therapy (beta-blockers, antiepileptics, antidepressants, CGRP inhibitors) is used for frequent or disabling attacks. Tension-type headaches respond to analgesics, stress management, and physical therapy. Cluster headaches may require high-flow oxygen, subcutaneous sumatriptan, and preventive agents (verapamil). Secondary headaches require treatment of the underlying cause (antibiotics for meningitis, urgent neurosurgical care for hemorrhage, corticosteroids for temporal arteritis). Avoid medication overuse and address triggers.

Heart Failure (Congestive Heart Failure) Symptoms Heart failure causes exertional dyspnea, orthopnea, paroxysmal nocturnal dyspnea, fatigue, reduced exercise tolerance, peripheral edema, and signs of fluid overload (jugular venous distension, pulmonary crackles). Right-sided failure produces hepatomegaly and ascites; left-sided failure leads to pulmonary congestion. Acute decompensation presents with severe dyspnea, hypoxia, and hypotension. **Treatment** Management includes diuretics for volume overload, ACE inhibitors or ARBs (or ARNI), beta-blockers, mineralocorticoid receptor antagonists, and SGLT2 inhibitors to improve symptoms and survival. Device therapy (ICD, CRT) and advanced options (inotropes, mechanical circulatory support, transplantation) are used for selected patients. Treat precipitating causes (ischemia, arrhythmia, infection), optimize comorbidities, and provide patient education on sodium/fluid restriction and self-monitoring.

Hepatitis (Viral and Nonviral Liver Inflammation) Symptoms Hepatitis presents with fatigue, anorexia, nausea, right upper-quadrant discomfort, jaundice, dark urine, pale stools, and elevated liver enzymes. Acute viral hepatitis (A, B, C, D, E) may be symptomatic or subclinical; chronic hepatitis (B, C) leads to progressive fibrosis, cirrhosis, and hepatocellular carcinoma risk. Alcoholic and nonalcoholic steatohepatitis cause similar features with metabolic risk factors. **Treatment** Acute supportive care for symptomatic viral hepatitis A/E; antiviral therapy for chronic

hepatitis B (nucleos(t)ide analogues) and hepatitis C (direct-acting antivirals) to achieve viral suppression or cure. Manage complications of chronic liver disease (portal hypertension, varices, encephalopathy) and counsel on alcohol cessation, weight loss for NAFLD, vaccination (A, B), and screening for hepatocellular carcinoma in high-risk patients.

HIV Infection (Human Immunodeficiency Virus) Symptoms Early HIV infection may be asymptomatic or present with a mononucleosis-like illness (fever, sore throat, lymphadenopathy, rash). Progressive untreated infection leads to immunosuppression, opportunistic infections, weight loss, chronic diarrhea, and AIDS-defining illnesses. Laboratory markers include declining CD4+ T-cell counts and rising viral load.

Treatment Antiretroviral therapy (ART) combining agents from different classes (integrase inhibitors, NRTIs, NNRTIs, protease inhibitors) suppresses viral replication, restores immune function, and reduces transmission. Treatment is individualized based on resistance testing and comorbidities. Preventive measures include PrEP for high-risk individuals and PEP after exposure. Manage opportunistic infections, provide vaccinations when appropriate, and offer psychosocial support and adherence counseling.

Hypertension (High Blood Pressure) Symptoms Hypertension is often asymptomatic; when present, symptoms are nonspecific—headache, dizziness, visual changes, or palpitations. Long-standing uncontrolled hypertension causes end-organ damage: left ventricular hypertrophy, coronary artery disease, stroke, chronic kidney disease, and retinopathy. Hypertensive emergencies present with acute end-organ dysfunction (encephalopathy, myocardial ischemia, renal failure). **Treatment** Lifestyle modifications (dietary sodium reduction, DASH diet, weight loss, exercise, alcohol moderation) and pharmacotherapy (thiazide diuretics, ACE inhibitors, ARBs, calcium channel blockers, beta-blockers) tailored to comorbidities. Target blood-pressure goals individualized by age and risk. In hypertensive emergencies, controlled IV antihypertensives in an ICU setting are required. Regular monitoring and adherence support reduce cardiovascular risk.

Hyperthyroidism and Hypothyroidism (Thyroid Disorders) Symptoms

Hyperthyroidism: weight loss, heat intolerance, palpitations, tremor, anxiety, increased appetite, and goiter; severe cases cause atrial fibrillation and thyrotoxic crisis.

Hypothyroidism: fatigue, weight gain, cold intolerance, constipation, dry skin, bradycardia, and cognitive slowing; severe hypothyroidism (myxedema) can be life-threatening. **Treatment** Hyperthyroidism: antithyroid drugs (methimazole, propylthiouracil), radioactive iodine ablation, or surgery for definitive therapy; beta-blockers for symptomatic control. Hypothyroidism: lifelong levothyroxine replacement with dose titration by TSH levels. Treat precipitating causes and monitor for complications (cardiac disease, osteoporosis).

Influenza (Seasonal Flu) Symptoms Influenza causes abrupt onset fever, myalgia, headache, cough, sore throat, and malaise; complications include pneumonia, exacerbation of chronic cardiopulmonary disease, and secondary bacterial infections, particularly in the elderly and immunocompromised. **Treatment** Antiviral agents (oseltamivir, zanamivir, baloxavir) are most effective when started early and reduce symptom duration and complications in high-risk patients. Supportive care includes hydration, antipyretics, and rest. Annual vaccination is the primary preventive strategy and reduces severe outcomes.

Ischemic Stroke and Transient Ischemic Attack (TIA) Symptoms Sudden focal neurologic deficits—weakness or numbness of face/arm/leg (usually unilateral), speech disturbance, visual loss, dizziness, or ataxia. TIAs produce transient deficits that resolve within 24 hours but signal high stroke risk. Rapid recognition and time-sensitive evaluation are critical. **Treatment** Acute ischemic stroke: urgent assessment for reperfusion therapy (IV thrombolysis within time window, mechanical thrombectomy for large-vessel occlusion) and supportive care. Secondary prevention includes antiplatelet or anticoagulant therapy (for cardioembolic sources), blood-pressure and lipid control, diabetes management, smoking cessation, and carotid intervention when indicated. Rehabilitation addresses functional recovery.

Irritable Bowel Syndrome (IBS) and Functional GI Disorders Symptoms IBS presents with recurrent abdominal pain associated with altered bowel habits (constipation, diarrhea, or mixed), bloating, and relief with defecation. Symptoms are chronic and fluctuate, often triggered by stress or certain foods. No structural disease is found on routine testing. **Treatment** Management is symptom-directed: dietary modification (low-FODMAP diet), fiber supplementation, antispasmodics, laxatives or antidiarrheals as needed, and psychological therapies (CBT, gut-directed hypnotherapy). Address comorbid anxiety or depression and provide patient education on chronicity and self-management.

Kidney Stones (Nephrolithiasis) Symptoms Renal colic presents with sudden, severe flank pain radiating to the groin, hematuria, nausea, and vomiting. Fever suggests superimposed infection and requires urgent attention. Stone size and location determine symptoms and passage likelihood. **Treatment** Pain control, hydration, and medical expulsive therapy (alpha-blockers) for small distal stones. Larger stones or obstructing stones with infection require urologic intervention—ureteroscopy with laser lithotripsy, shock-wave lithotripsy, or percutaneous nephrolithotomy. Preventive measures include metabolic evaluation, dietary modification, and medications (thiazides, citrate) tailored to stone composition.

Liver Cancer (Hepatocellular Carcinoma) Symptoms Often asymptomatic early; later signs include weight loss, abdominal pain, hepatomegaly, jaundice, ascites, and decompensation in cirrhotic patients. Screening in high-risk populations (cirrhosis,

chronic hepatitis B/C) detects tumors earlier. **Treatment** Curative options for early disease include surgical resection, liver transplantation, or ablative therapies. Intermediate and advanced disease may be treated with transarterial chemoembolization, targeted systemic therapies (tyrosine kinase inhibitors), and immunotherapy. Management includes surveillance, treatment of underlying liver disease, and palliative care when needed.

Lung Cancer Symptoms Persistent cough, hemoptysis, dyspnea, chest pain, unexplained weight loss, and paraneoplastic syndromes. Many cases are detected at advanced stages; smoking is the primary risk factor. **Treatment** Management depends on histology and stage: surgical resection for early disease, radiation therapy, systemic chemotherapy, targeted therapies for driver mutations (EGFR, ALK, ROS1), and immunotherapy for advanced disease. Smoking cessation, screening in high-risk individuals, and multidisciplinary care improve outcomes.

Lyme Disease (Borrelia burgdorferi Infection) Symptoms Early localized disease: erythema migrans rash (expanding target lesion), fever, fatigue, headache, and myalgias. Disseminated disease causes multiple rashes, neurologic involvement (facial palsy, meningitis), and cardiac conduction abnormalities. Late disease may produce arthritis and neurologic symptoms. **Treatment** Early disease: oral doxycycline (or amoxicillin/cefuroxime in children/pregnancy) for 10–21 days. Disseminated or neurologic disease may require IV ceftriaxone. Symptom management and follow-up for persistent symptoms are individualized; prevention includes tick avoidance and prompt removal.

Malaria Symptoms Cyclical fevers, chills, sweats, headache, myalgias, malaise, and in severe cases anemia, jaundice, respiratory distress, cerebral involvement, and multiorgan failure. Travel history to endemic areas is key to diagnosis. **Treatment** Species-specific antimalarial therapy (artemisinin-based combination therapy for P. falciparum in many regions; chloroquine for sensitive strains). Severe malaria requires parenteral therapy (IV artesunate) and intensive supportive care. Prevention includes chemoprophylaxis for travelers, insecticide-treated bed nets, and vector control.

Melanoma and Skin Cancer Symptoms New or changing pigmented lesion, asymmetry, border irregularity, color variation, diameter >6 mm, or evolution (ABCDE). Nonmelanoma skin cancers (basal cell, squamous cell) present as nonhealing nodules or ulcers. Early detection improves prognosis. **Treatment** Surgical excision with appropriate margins is primary therapy; sentinel lymph node biopsy for staging in melanoma. Advanced melanoma may be treated with immunotherapy (checkpoint inhibitors) and targeted agents for BRAF-mutant tumors. Prevention includes sun protection and skin surveillance.

Multiple Sclerosis (MS) Symptoms Relapsing-remitting or progressive neurologic deficits—optic neuritis, limb weakness, sensory disturbances, ataxia, bladder dysfunction, and cognitive changes—reflecting demyelinating lesions in the CNS. Symptoms often fluctuate and accumulate over time. **Treatment** Acute relapses treated with high-dose corticosteroids; disease-modifying therapies (interferons, glatiramer acetate, monoclonal antibodies, oral agents) reduce relapse rate and slow progression. Symptomatic therapies, rehabilitation, and multidisciplinary care address spasticity, fatigue, bladder dysfunction, and mood disorders.

Myocardial Infarction (Heart Attack) Symptoms Chest pain or pressure often radiating to jaw, neck, shoulder, or arm, accompanied by diaphoresis, nausea, dyspnea, and sometimes syncope. Atypical presentations occur in women, elderly, and diabetics. ECG changes and cardiac biomarkers confirm diagnosis. **Treatment** Immediate reperfusion therapy (primary PCI preferred; thrombolysis when PCI unavailable), antiplatelet agents, anticoagulation, analgesia, oxygen if hypoxic, and beta-blockers. Post-MI care includes statins, ACE inhibitors, cardiac rehabilitation, and secondary prevention measures.

Obesity (Overweight and Related Metabolic Disease) Symptoms Excess adiposity with increased BMI, often accompanied by metabolic complications—insulin resistance, dyslipidemia, hypertension, obstructive sleep apnea, and increased cardiovascular risk. Psychological and functional impairments are common. **Treatment** Lifestyle interventions (dietary change, increased physical activity, behavioral therapy) are first-line. Pharmacotherapy (GLP-1 receptor agonists, other agents) and bariatric surgery are options for selected patients with severe obesity or comorbidities. Multidisciplinary care addresses nutrition, exercise, psychological support, and long-term follow-up.

Osteoporosis and Osteopenia Symptoms Often asymptomatic until fragility fractures occur (vertebral compression, hip, wrist), causing pain, height loss, kyphosis, and disability. Risk factors include age, female sex, low body weight, smoking, glucocorticoid use, and family history. **Treatment** Bone-density testing guides therapy. Calcium and vitamin D supplementation, weight-bearing exercise, fall prevention, and pharmacologic therapy (bisphosphonates, denosumab, selective estrogen receptor modulators, parathyroid hormone analogs) reduce fracture risk. Address secondary causes and monitor therapy.

Osteoarthritis (Degenerative Joint Disease) Symptoms Joint pain worsened by use, stiffness after inactivity, crepitus, and progressive loss of function—commonly affecting knees, hips, hands, and spine. Deformity and reduced mobility occur in advanced disease. **Treatment** Nonpharmacologic measures (exercise, weight loss, physical therapy) are foundational. Analgesics (acetaminophen, NSAIDs), topical agents,

intraarticular corticosteroids, and joint replacement surgery for end-stage disease restore function and relieve pain.

Pancreatitis (Acute and Chronic) Symptoms Acute pancreatitis presents with severe epigastric pain radiating to the back, nausea, vomiting, and elevated pancreatic enzymes; severe cases cause systemic inflammatory response and organ failure. Chronic pancreatitis causes persistent pain, malabsorption, steatorrhea, and diabetes.

Treatment Acute: aggressive IV fluids, pain control, bowel rest, and treatment of underlying causes (gallstones, alcohol). Severe cases require ICU care and management of complications (necrosis, infection). Chronic: alcohol cessation, pancreatic enzyme replacement, pain management, endoscopic or surgical interventions for ductal obstruction, and nutritional support.

Parkinson Disease Symptoms Progressive bradykinesia, resting tremor, rigidity, postural instability, and gait disturbance; nonmotor symptoms include autonomic dysfunction, sleep disorders, mood changes, and cognitive decline. **Treatment** Levodopa remains the most effective symptomatic therapy; dopamine agonists, MAO-B inhibitors, and COMT inhibitors are adjuncts. Deep brain stimulation is an option for medication-refractory motor complications. Multidisciplinary care addresses physical therapy, speech therapy, and management of nonmotor symptoms.

Pneumonia (Community-Acquired and Hospital-Acquired) Symptoms Fever, productive cough, pleuritic chest pain, dyspnea, and crackles on auscultation; elderly patients may present with confusion or falls. Chest imaging shows lobar or patchy infiltrates. **Treatment** Empiric antibiotics guided by severity, setting (community vs hospital), and risk factors; supportive care includes oxygen, fluids, and respiratory therapy. Vaccination (influenza, pneumococcal) reduces risk. Hospitalization for severe disease and ICU care for respiratory failure or sepsis.

Prostate Cancer Symptoms Early prostate cancer is often asymptomatic; later symptoms include urinary hesitancy, weak stream, hematuria, bone pain from metastases, and elevated PSA. Screening and shared decision-making guide detection. **Treatment** Options include active surveillance for low-risk disease, radical prostatectomy, radiation therapy, androgen-deprivation therapy for advanced disease, and systemic therapies (chemotherapy, novel hormonal agents) for metastatic disease. Manage treatment side effects and provide survivorship care.

Psoriasis Symptoms Chronic immune-mediated skin disease with well-demarcated erythematous plaques and silvery scale, commonly on extensor surfaces and scalp; may be associated with psoriatic arthritis, nail changes, and systemic inflammation. **Treatment** Topical therapies (corticosteroids, vitamin D analogues), phototherapy, systemic agents (methotrexate, cyclosporine), and biologic therapies (TNF inhibitors,

IL-17/IL-23 inhibitors) for moderate-to-severe disease. Address comorbidities (cardiometabolic risk) and provide patient education.

Sepsis and Septic Shock Symptoms Life-threatening organ dysfunction caused by a dysregulated host response to infection—fever or hypothermia, tachycardia, hypotension, altered mental status, oliguria, and laboratory evidence of organ dysfunction. Septic shock includes persistent hypotension requiring vasopressors and elevated lactate. **Treatment** Early recognition and rapid initiation of broad-spectrum IV antibiotics, source control (drainage, debridement), aggressive fluid resuscitation, vasopressors for refractory hypotension, and organ support in ICU. Adherence to sepsis bundles and timely escalation improve outcomes.

Sexually Transmitted Infections (STIs) Symptoms STIs produce a range of genital and systemic symptoms—urethral or vaginal discharge, genital ulcers, pelvic pain, dysuria, and systemic signs in disseminated infections. Many infections (chlamydia, gonorrhea, HPV, HIV, syphilis) may be asymptomatic. **Treatment** Diagnosis by testing guides therapy: antibiotics for bacterial STIs (azithromycin/doxycycline for chlamydia, ceftriaxone for gonorrhea, penicillin for syphilis), antivirals for herpes, and counseling, partner notification, and prevention strategies (condoms, vaccination for HPV and hepatitis B). Routine screening in at-risk populations reduces transmission.

Stroke (Ischemic and Hemorrhagic) — *See Ischemic Stroke entry above for ischemic management; hemorrhagic stroke requires neurosurgical evaluation, blood-pressure control, reversal of anticoagulation when applicable, and intensive supportive care.*

Systemic Lupus Erythematosus (SLE) Symptoms Multisystem autoimmune disease with variable manifestations—malar rash, photosensitivity, arthritis, serositis, renal involvement (nephritis), hematologic abnormalities, and neurologic symptoms. Flares and remissions are typical. **Treatment** Tailored to organ involvement: hydroxychloroquine for most patients, NSAIDs and low-dose corticosteroids for mild disease, and immunosuppressive agents (azathioprine, mycophenolate, cyclophosphamide, biologics) for severe organ-threatening disease. Sun protection, vaccination, and monitoring for complications are important.

Thyroid Cancer Symptoms Often presents as a painless thyroid nodule; some patients have hoarseness or compressive symptoms. Most thyroid cancers (papillary, follicular) have favorable prognosis; medullary and anaplastic types are more aggressive.

Treatment Surgical thyroidectomy with or without radioactive iodine ablation for differentiated cancers; TSH suppression therapy and long-term surveillance with thyroglobulin monitoring. Multidisciplinary care for advanced or aggressive histologies.

Tuberculosis (Pulmonary and Extrapulmonary TB) Symptoms Pulmonary TB: chronic cough, hemoptysis, night sweats, weight loss, and fever. Extrapulmonary TB affects lymph nodes, pleura, bones, CNS, and genitourinary tract with organ-specific signs.

Risk factors include immunosuppression and close contact with infectious cases.

Treatment Multi-drug antitubercular therapy (isoniazid, rifampin, pyrazinamide, ethambutol) for active disease with adherence support and monitoring for toxicity. Latent TB infection is treated to prevent progression. Public-health measures include contact tracing and isolation of infectious cases.

Ulcerative Colitis (Inflammatory Bowel Disease) Symptoms Continuous mucosal inflammation of the colon causing bloody diarrhea, urgency, abdominal pain, and tenesmus. Severe flares can cause systemic toxicity and risk of toxic megacolon.

Extraintestinal manifestations include arthritis and skin lesions. **Treatment** Aminosalicylates for mild disease, corticosteroids for induction of remission, immunomodulators and biologic agents for moderate-to-severe or steroid-dependent disease, and colectomy for refractory disease or dysplasia. Surveillance colonoscopy for cancer risk in long-standing disease.

Urinary Tract Infection (Cystitis, Pyelonephritis) Symptoms Cystitis: dysuria, urinary frequency, urgency, suprapubic discomfort, and sometimes hematuria. Pyelonephritis: fever, flank pain, nausea, vomiting, and systemic signs. Elderly patients may present atypically with confusion. **Treatment** Uncomplicated cystitis: short-course oral antibiotics guided by local resistance patterns and symptomatic care. Pyelonephritis and complicated infections require longer courses and sometimes IV antibiotics and hospitalization. Recurrent UTIs prompt evaluation for predisposing factors and preventive strategies.

Viral Hepatitis — See *Hepatitis entry above for viral and nonviral causes and treatment approaches.*

Vertigo and Vestibular Disorders Symptoms Vertigo is the sensation of spinning or motion, often accompanied by nausea, vomiting, imbalance, and nystagmus. Peripheral causes (benign paroxysmal positional vertigo, vestibular neuritis, Meniere disease) differ from central causes (stroke, multiple sclerosis) by associated neurologic signs. **Treatment** Peripheral vertigo: repositioning maneuvers (Epley) for BPPV, vestibular suppressants short-term, and vestibular rehabilitation. Meniere disease managed with salt restriction, diuretics, intratympanic steroids, or surgical options for refractory cases. Central vertigo requires treatment of the underlying neurologic disorder.

Viral Infections (General: Dengue, Zika, COVID-19, etc.) Symptoms Viral infections produce a spectrum from mild febrile illness to severe systemic disease depending on the virus—fever, myalgias, respiratory or gastrointestinal symptoms, rash, and organ-specific complications. Emerging viruses (SARS-CoV-2) can cause severe respiratory failure and multisystem involvement. **Treatment** Supportive care is central for many viral illnesses; specific antivirals exist for some (oseltamivir for influenza,

direct-acting antivirals for hepatitis C, remdesivir for selected COVID-19 cases). Vaccination, vector control, and public-health measures prevent spread. Severe cases require organ support and targeted therapies when available.

Wound Care and Infection Prevention Symptoms Wounds range from superficial abrasions to deep lacerations and puncture injuries; signs of infection include increasing pain, erythema, warmth, purulent drainage, and systemic fever. Chronic wounds (venous stasis, diabetic foot ulcers) show delayed healing and risk of osteomyelitis. **Treatment** Initial wound care includes cleansing, debridement of devitalized tissue, appropriate closure when indicated, tetanus prophylaxis, and infection prevention. Infected wounds require targeted antibiotics and sometimes surgical drainage. Chronic wound management uses offloading, vascular assessment, infection control, and advanced dressings or grafting as needed.

Zika Virus Infection Symptoms Zika infection is often mild with fever, rash, conjunctivitis, arthralgia, and headache. Infection during pregnancy can cause congenital Zika syndrome with microcephaly and severe neurologic impairment in the fetus. Guillain-Barré syndrome is a rare neurologic complication. **Treatment** Supportive care—rest, fluids, analgesics, and antipyretics. Prevention focuses on mosquito control, travel advisories for pregnant women, and sexual transmission precautions. Pregnant women with exposure requ